Metal-pad-roll instability theory for small-scale models of reduction cells

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We present a theoretical model for the metal pad roll instability in laboratory-scale reduction cell experimental models consisting of two stably stratified liquid layers subjected to a vertical electric current and a likewise vertical magnetic field. In contrast to most previous studies, we do not rely on the shallow-water approximation and can account for viscous and capillary effects that are prevalent in small-scale experiments. In small cells, dissipation effects are largely attributable to laminar viscous damping, which we calculate analytically using a perturbative formulation of the Stokes boundary layers forming at all container walls and on both sides of the liquid interface. The derived damping rates, which are also useful for modeling liquid-liquid sloshing in rectangular containers, are validated against direct numerical simulations conducted with the coupled level set and volume-of-fluid solver ARCHER. On this basis, we derive analytical solutions for the growth rates and stability onsets of the metal pad roll instability, which we compare with different theoretical descriptions and existing experiments. The presented solutions are intended to serve as a foundation for future benchmarking of multiphase solvers.

I. INTRODUCTION

Over the past 50 years, the modeling of the metal pad roll (MPR) instability has emerged as a flagship application in the discipline of magnetohydrodynamics, and it can be found in textbooks today [1, 2]. The high level of interest is attributable to the practical significance for aluminum production. Aluminum reduction cells (ARCs) are known to be vulnerable to the MPR instability, and physical modeling can lead to strategies that help in keeping the instability under control. In terms of hydrodynamics, ARCs can be regarded as stably stratified systems of two liquid layers carrying a strong vertical electric current, see Fig. 1(a). The alumina is dissolved in molten cryolite (Na_3AlF_6) and floats on top of a layer of reduced pure aluminum. The interface between both layers can be destabilized and set in motion by Lorentz forces, interactions between the electrical currents in the cell, and induced or external magnetic fields. Many studies have shown that the most critical instability results from the interaction of horizontal electrical currents in the aluminum layer and the vertical component of an external magnetic field (B_z in the figure) [2, chapter 6.1]. As illustrated in Fig. 1(b), external magnetic fields are induced by the power supply lines in the aluminium factory and cannot be avoided during operation. In this paper, we use the term MPR instability to refer to this

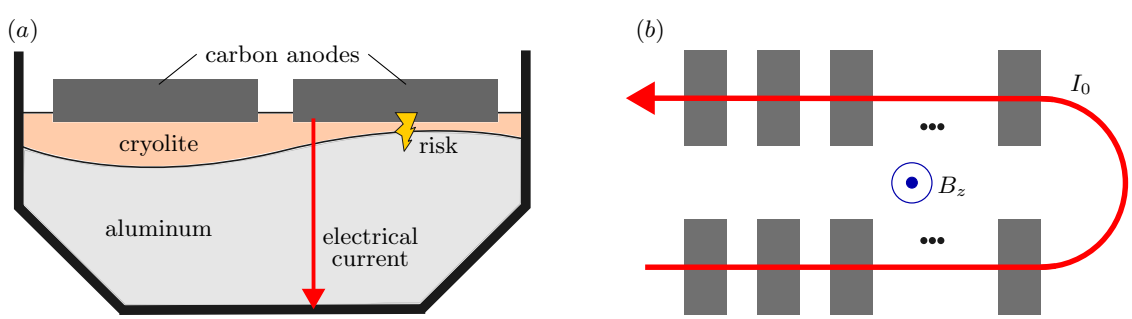

FIG. 1. (a) Sideview of an aluminium reduction cell (ARC) with two superposed liquid layers. Metal pad roll instability destabilizes waves on the interface and can cause short-circuits. (b) ARC's are typically arranged around a central current loop in an industrial hall and this creates an ambient magnetic field B_z of typically a few mT.

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particular scenario, in line with the majority of the literature. The MPR instability causes a rotating sloshing motion in the aluminum (metal pad) that must be strictly controlled throughout the electrolysis. As indicated in Fig. 1(a), the cells can be short-circuited if the liquid aluminum layer makes contact with the upper carbon anodes. For ARCs to remain stable with respect to the MPR-instability, the cryolite layer must not be thinner than about 3.5–4.5 cm [3]. This is, however, very unfavorable energetically as important Ohmic losses occur in the poorly conducting cryolite. About 40 % of a cell's electrical energy reduces no aluminum, but is converted into waste heat. Keeping in mind that aluminum production consumes around 8 % of the global industrial electricity [4], the importance of understanding and controlling MPR instability is evident. Recently, there has also been interest regarding the fundamental aspects of the instability in context to liquid metal batteries (LMBs): a promising technology for grid-scale energy storage applications, intended to accommodate fluctuations in the supply of energy from renewable sources. Flow instabilities occurring inside this molten metal-salt system can have a crucial impact on the cell's efficiency and operational safety [5–7].

The first successful attempts to explain the MPR instability mechanism go back to the studies of Urata [8] and Sele [9]. On the basis of some heuristic arguments, Sele succeeded in deducing an essential non-dimensional parameter that controls the stability

$$\beta_{\mathrm{S}} = \frac{I_0 B_z}{(\rho_{\mathrm{a}} - \rho_{\mathrm{c}}) g h_{\mathrm{a}} h_{\mathrm{c}}}, \tag{1}$$

today referred to as the Sele parameter. In this parameter, I_0 and B_z denote the total cell current and the vertical magnetic field, g is gravity and $\rho_{\mathrm{a}}, \rho_{\mathrm{c}}$ and $h_{\mathrm{a}}, h_{\mathrm{c}}$ refer to the densities and heights of the aluminum and cryolite layers. According to Sele, ARCs are MPR unstable when β_{S} exceeds a critical value, β_{crit}, which depends on the lateral cell geometry, but also on the viscous or turbulent damping. The scaling behavior $\beta_{\mathrm{S}} \sim 1/h_{\mathrm{c}}$ explains why we need a thick enough cryolite layer.

Much progress in the theoretical modelling of the MPR instability was made in the 1990s [10–12]. Perturbation techniques were applied to study the linear stability in the framework of the shallow-water approximation. It was revealed that the Lorentz force can couple different standing gravity wave modes and destabilise them by pairs, much like in a parametric resonance of waves. The MPR instability is triggered when the natural frequencies ω and ω' of at least two transverse gravity wave modes are sufficiently close to one another. The effect of the magnetohydrodynamic coupling is that ω and ω' are shifted towards each other for increasing $I_0 B_z$ until they coincide at $\beta_{\mathrm{S}} = \beta_{\mathrm{crit}}$. If $I_0 B_z$ is further increased ($\beta_{\mathrm{S}} > \beta_{\mathrm{crit}}$), the wave-pair becomes unstable and grows exponentially to become a rotating metal pad wave. The stability threshold β_{crit} scales with the difference between the squared eigenfrequencies $\beta_{\mathrm{crit}} \sim |\omega^2 - \omega'^2|$ [12]. Cells in which pairs of gravity waves with identical natural frequencies $\omega = \omega'$ exist (e.g., this is the case for square and cylindrical cell geometries) are always unstable in the inviscid limit; the stability threshold in such cells only depends on the viscous or turbulent damping. An alternative view on the origin of the MPR instability was given by Lukyanov *et al.* [13], who argued that the instability is the result of multiple self-amplifying wave reflections on the sidewalls. According to Lukyanov *et al.* [13], the MHD coupling mechanism is inherently connected to the lateral walls, explaining the vocabulary of *wall modes* introduced by Molokov *et al.* [14]. In the decade that followed, most attention went to the development of nonlinear numerical models [2, 15–17] that can describe wave dynamics well beyond the stability threshold, e.g., for estimating saturation amplitudes. This has allowed ARC modeling to reach a certain level of maturity, and a great deal of effort has gone into the optimization of magnetic field and electric current distributions [18] to maximally avoid the MPR-instability.

Conducting well-controlled experiments on MPR instability in ARCs and LMBs is difficult due to the high temperature ($\sim 1000^o$C) and chemical aggressiveness. For this reason, several attempts have been made to develop downscaled model experiments that can synthesize or imitate MPR waves under ambient conditions. The pioneering experiment was devised by Pedcenko *et al.* [19, 20], who replaced the cryolite layer with a grid of vertical thin rods imitating its electrical role. The aluminum layer was replaced by the eutectic alloy GaInSn that is liquid at room temperature. The experiment succeeded in demonstrating the MPR instability under a high imposed magnetic field, but also suffered some limitations. The steel rods were quite intrusive, and the instability was artificially intensified when the GaInSn detached from them under wave motion. This experiment can also not be employed as a benchmark for MPR theories that suppose two fluid layers. At this point, we would like to take the opportunity to highlight that another very sophisticated MPR experiment has been done using two immiscible transparent solutions of nitric acid ($HNO_{3(\mathrm{aq})}$ at the bottom and $HNO_3 + C_5H_{12}O$ at the top). Unfortunately, this work of Borisov *et al.* [21] has only been published in a Russian-language journal, and therefore it has been overlooked. In this set-up, both liquids with very similar densities were used to destabilize the interface with small currents in the presence of strong magnetic fields. This is the only study that has ever reported experimental stability onsets, including the associated most unstable wave pairs as a function of the lateral aspect ratio of the cell. A more realistic MPR experiment setup using liquid Gallium and aqueous electrolyte was conducted by Grants and Baranovskis [22], who observed a sustained rotating surface wave instead of a growing interfacial wave. The authors demonstrated that the horizontal rotation of the liquids

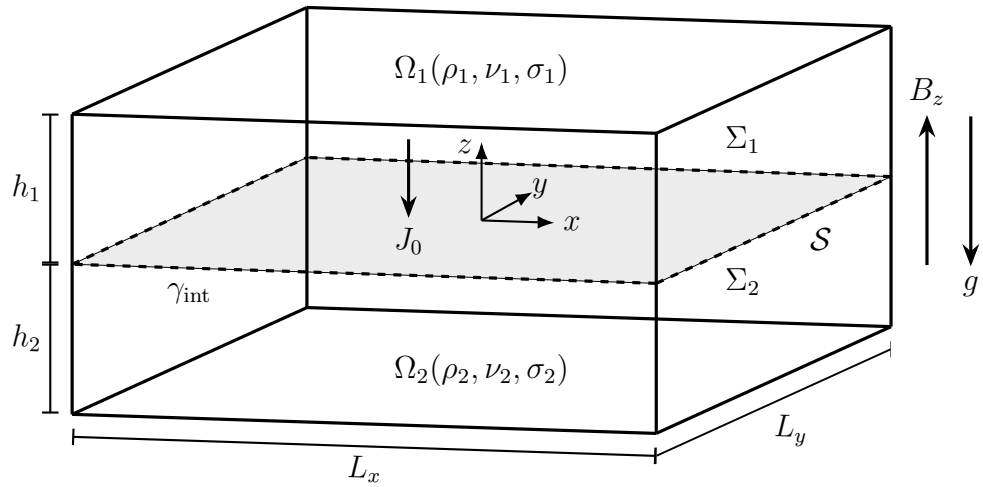

FIG. 2. Schematic of the cell at equilibrium consisting of two fluid mediums Ω_1 (upper), Ω_2(lower) characterized by their densities ρ_1, ρ_2, kinematic viscosities ν_1, ν_2, electrical conductivity σ_1, σ_2, thicknesses h_1,h_2 and bounded by rigid boundaries (container walls) Σ_1,Σ_2 respectively. Gravity g acts in the negative z direction, stratifying the two media with an interface $\mathcal{S}$ located at $z = 0$ and characterized by interfacial tension γ_{int}. Both layers are traversed by a homogeneous vertical electrical current density J_0 and submitted to a homogeneous vertical magnetic field B_z.

significantly delayed the onset of the MPR instability. A similar two-layer setup using Ga-Hg liquid metal pairs was proposed [23], but no experiment has been done yet.

Recently, Hegde *et al.* [24] have proposed a new small-scale MPR experiment that works with one liquid layer of GaInSn. The MPR instability is synthesized through a feedback loop control that injects horizontal electrical currents in the GaInSn-layer, proportional to the measured slope. This imitates the physics of the Sele-mechanism, but the feedback mechanism still needs to be carefully calibrated to imitate the MPR physics of two-layer cells. Another recent experiment is that of Eltishchev *et al.* [25, 26], who destabilized rotating surface waves through a MHD forcing mechanism that is quite similar to that of the MPR instability.

Spurred on by these recent experimental advances and the rediscovery of Borisov's experiment, we present here a linear stability theory that is specially dedicated to MPR instability in small-scale experimental models. In small-scale set-ups, we must account for physical ingredients that are usually neglected in ARCs, and this requires a specific theory. We can have non-shallow layers, significant viscous damping, possibly magnetic damping (for high B_z), and low differences in electrical conductivity between the two liquid layers. Herreman *et al.* [27] proposed a linear stability theory for small cylindrical cells, whereas we present here a similar theory for small, rectangular cells. A perturbative approach was also used in Politis and Priede [28], but wave-damping there was modeled in a phenomenological way. In this study, we propose a laminar viscous damping model that has no adjustable parameters and can also be compared to previous work on viscous damping of waves in two-layer systems, [29] and [30]. A linear stability model consisting of equations for the growth rate and the viscous damping rate will be essential in benchmarking numerical solvers and existing MPR experiments, as well as for developing new ones.

The manuscript is organized as follows: In Sec. II we calculate the growth rate of the MPR instability in small cells using perturbation methods and including a model for viscous dissipation. In Sec. III, we validate our new two-layer damping formula numerically with the DNS multiphase code ARCHER, and we also compare our damping rates to the experimental and numerical results of La Rocca *et al.* [30]. We then calculate theoretically predicted onsets of stability as a function of cell geometry and compare with the theory of Politis and Priede [28]. We also apply our stability theory to the two-layer experimental set-up of Borisov *et al.* [21]. In Sec. IV we collate our results and discuss open questions.

II. LINEAR STABILITY ANALYSIS

A. Formulation of the problem

We consider a system of two superposed immiscible, incompressible and Newtonian liquid layers (Ω_1 and Ω_2) at equilibrium in a rectangular domain (x, y, z) of dimensions L_x, L_y and $H = h_1 + h_2$, as shown in Fig. 2. For both layers, the density ρ_i, kinematic viscosity ν_i and electrical conductivity σ_i are taken into account. The subscripts

$i = 1, 2$ refer to the upper and lower layers respectively. The origin of the coordinate system $\mathcal{O}$ is located at the center of the interface. The rigid boundaries of each layer are denoted by Σ_i. We include interfacial tension γ_{int} to our model, which acts in addition to gravity g as a restoring force, but neglect capillary effects at the side walls, i.e., we assume that the contact line slides freely along the side walls while maintaining a static contact angle of 90°. The entire cell is exposed to an external homogeneous magnetic field oriented vertically B_z, and gravity acts in the negative z direction $\boldsymbol{g} = -g\boldsymbol{e}_z$. At equilibrium, both layers carry a homogeneous vertical electrical current density $J_0 = -I_0/(L_x L_y)$, where I_0 denotes the total cell current. The equilibrium configuration is fully characterized by the hydrodynamic pressure $P_i = P_0 - \rho_i g z$, and the electrical potential $\Phi_i = \Phi_0 - J_0 z/\sigma_i$, where P_0 and Φ_0 are constants. In the perturbed state, the total current density $\boldsymbol{J}(x, y, z, t) = J_0\boldsymbol{e}_z + \boldsymbol{j}_i(x, y, z, t)$ takes into account the excess currents $\boldsymbol{j}_i$ and the total magnetic field strength $\boldsymbol{B}(x, y, z, t) = B_z\boldsymbol{e}_z + \boldsymbol{b}_i(x, y, z, t)$ takes into account the induced magnetic fields $\boldsymbol{b}_i$. The perturbed interface $\mathcal{S}$ is located at $z = \eta(x, y, t)$.

B. Governing equations and boundary conditions

The quantities $(\boldsymbol{u}_i, p_i, \boldsymbol{j}_i, \varphi_i, \boldsymbol{b}_i)$, for the perturbations of velocity, pressure, current density, electrical potential, and induced magnetic field which modify the equilibrium state [27, 31], are governed by the following equations

$$\rho_i \partial_t \boldsymbol{u}_i + \nabla p_i = \boldsymbol{j}_i \times \boldsymbol{B} + \rho_i \nu_i \nabla^2 \boldsymbol{u}_i, \tag{2a}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{u}_i = 0, \tag{2b}$$

$$\boldsymbol{\nabla} \cdot \boldsymbol{j}_i = 0, \tag{2c}$$

$$\boldsymbol{j}_i = -\sigma_i(\boldsymbol{\nabla}\varphi_i + \boldsymbol{u}_i \times \boldsymbol{B}), \tag{2d}$$

$$\boldsymbol{\nabla} \times \boldsymbol{j}_i = -\frac{\partial \boldsymbol{b}_i}{\partial t}, \tag{2e}$$

$$\boldsymbol{\nabla} \times \boldsymbol{b}_i = \mu_0 \boldsymbol{j}_i, \qquad \boldsymbol{\nabla} \cdot \boldsymbol{b}_i = 0. \tag{2f}$$

Eq. (2a) is the linearized conservation of momentum relation; Eq. (2b) expresses the continuity of velocity and (2c) the excess current. Eq. (2d) refers to the quasi-magnetostatic approximation of Ohm's law [1]. Eq. (2d), Eq. (2e) (Faraday's law) and Eq. (2f)(Gauss's law) together give the following induction equation:

$$\frac{\partial \boldsymbol{b}_i}{\partial t} - \boldsymbol{\nabla} \times (\boldsymbol{u}_i \times \boldsymbol{B}) - \frac{\Delta \boldsymbol{b}_i}{\mu_0 \sigma_i}, \tag{3}$$

where $\mu_0\sigma_i$ is the magnetic diffusivity of a given material (μ_0 is the permeability of free space). Eq. (3) describes the advective and diffusive processes of the magnetic field in a given medium. To ascertain the limits of our theoretical model under the magnetostatic approximation, we introduce dimensionless quantities to characterize the system, mainly the Reynolds number Re, and the magnetic Reynolds number $\mathrm{R_m}$,

$$\mathrm{Re} = \omega L^2/\nu, \quad \mathrm{R_m} = \mu_0 \sigma \omega L^2. \tag{4}$$

The characteristic velocity can be written as a product of the interfacial wave frequency ω and the characteristic length of the geometry $L = \sqrt{L_x L_y}$. The magnetic Reynolds number $\mathrm{R_m}$ can be expressed as the ratio of the magnetic diffusion time $\tau_{\mathrm{m}} = \mu_0 \sigma L^2$, and the flow time $\tau_{\mathrm{f}} = \omega^{-1}$. At low $\mathrm{R_m}$, magnetic diffusion is much faster than fluid advection, so the field achieves equilibrium instantaneously relative to the global time scale. Therefore, we can neglect the terms $\partial \boldsymbol{b}_i/\partial t$ in Eq. 3. As $\boldsymbol{b}_i$ scales with $\mathrm{R_m}$, they ensure that the induced fields are comparatively small, thereby reducing $\boldsymbol{u} \times \boldsymbol{B} \approx 0$. Previous studies based on shallow water theories have largely ignored the contribution of the induced magnetic field on the perturbed Lorentz force [11, 12]. In a cylindrical system, it was shown that $\boldsymbol{J} \times \boldsymbol{b}_i$ does not contribute to the amplitude of the wave but rather to the frequency shift [27]. This is because the Lorentz force due to the horizontal component of the induced magnetic field $\boldsymbol{J} \times \boldsymbol{b}_i$ is out of phase with $\boldsymbol{J} \times B_z\boldsymbol{e}_z$ (the dominant contribution) and therefore, cannot inject power into the wave. These assumptions effectively, reduce the magnetic problem to a static case in both the liquid layers. Therefore, the following conditions: $\partial \boldsymbol{b}_i/\partial t \approx 0$, $\boldsymbol{u} \times \boldsymbol{B} \approx 0$ and $\boldsymbol{J} \times \boldsymbol{b}_i \approx 0$ allow us to impose the magnetostatic approximation on the system where the electric field is a dependent variable, only of the electric potential.

For the formulation of the boundary conditions at the tank walls, we account for the fact that the upper carbon anode in ARCs is a much better electrical conductor than the cryolite, and the bottom steel cathode often a much

poorer conductor than aluminum. For this idealized case, in which excess currents close entirely in the aluminum layer, we obtain the following linearized boundary conditions:

$$\boldsymbol{j}_1 \cdot \boldsymbol{e}_x\big|_{x=-L_x/2} = \boldsymbol{j}_2 \cdot \boldsymbol{e}_x\big|_{x=L_x/2} = 0, \tag{5a}$$

$$\boldsymbol{j}_1 \cdot \boldsymbol{e}_y\big|_{y=-L_y/2} = \boldsymbol{j}_2 \cdot \boldsymbol{e}_y\big|_{y=L_y/2} = 0, \tag{5b}$$

$$\varphi_1 = 0\,\big|_{z=h_1}, \tag{5c}$$

$$\boldsymbol{j}_2 \cdot \boldsymbol{e}_z = 0\,\big|_{z=-h_2}. \tag{5d}$$

Eqs. (5a) and (5b) ensure that no current can escape through the insulating side walls. The anode is modeled as an iso-potential surface [Eq. (5c)] and the weakly conducting cathode is treated as a perfect insulator [Eq. (5d)] for the excess current component $\boldsymbol{j}$. In Sec. III D, we further consider a case where the cathode is modeled as a perfect conductor, in that respect, condition [Eq. (5d)] is not valid for all reduction cells and MPR model experiments. As for the kinematic boundary conditions, we simply apply impermeability at all walls

$$\boldsymbol{u}_i \cdot \boldsymbol{n}_i|_{\Sigma_i} = 0, \tag{6}$$

for the inviscid formulation (no slip condition) in Sec. II C to which we must add

$$\boldsymbol{u}_i \times \boldsymbol{n}_i|_{\Sigma_i} = 0. \tag{7}$$

to account for viscous effects, we address in Sec. II E. At the interface $\mathcal{S}$, we have to take into account the following linearized hydrodynamic and electrical boundary conditions:

$$\partial_t \eta = u_{1,z}\,\big|_{z=0} = u_{2,z}\big|_{z=0}, \tag{8a}$$

$$-\gamma_{\text{int}}\nabla^2\eta + (\rho_2 - \rho_1)g\eta = p_2\,\big|_{z=0} - p_1\,\big|_{z=0}, \tag{8b}$$

$$\boldsymbol{j}_1 \cdot \boldsymbol{e}_z = \boldsymbol{j}_2 \cdot \boldsymbol{e}_z\,\big|_{z=0}, \tag{8c}$$

$$J_0(\sigma_1 - \sigma_2)\eta = \varphi_2\,\big|_{z=0} - \varphi_1\,\big|_{z=0}. \tag{8d}$$

Eqs. (8a) and (8b) represent the classic kinematic and dynamic boundary conditions for gravity-capillary waves. In Sec. II E, we state the corrections to Eq.(8b) incorporating the continuity of the tangential velocities and viscous stresses in order to solve the Stokes velocities (for viscous damping). The electrical boundary conditions are derived from the basic principle that both the normal of the total current density $\boldsymbol{n} \cdot \boldsymbol{J}$ and the total electric potential $(\Phi_i + \varphi_i)$ must remain continuous. This results in the linearized interfacial conditions [Eq. (8c)] and [Eq. (8d)] for the perturbed quantities [32]. This jump condition is essential for the MPR instability as it describes how the current density redistributes in both layers as a response to an arbitrary (small) interface elevation η. Because the horizontal component of the redistributed current is responsible for the Lorentz force that reinforces interface displacements η, the self-destabilization is mainly anchored in this interfacial boundary condition.

C. Two-wave perturbation approach

The perturbative approach is based on the (inviscid) potential flow solutions of free gravity-capillary interfacial waves that form a set of orthogonal functions in the absence of MHD effects ($\boldsymbol{B}, \boldsymbol{J} = \boldsymbol{0}$) and without viscosity ($\nu = 0$). Viscosity only affects the flow in the close vicinity of the Stokes boundary layers forming at the container walls, and below and above the interface. In Sec. II E, we calculate Stokes boundary layers corrections to the potential flow solutions to predict viscous damping rates associated with interfacial wave motions. In our formulation, however, only perturbations of the inviscid bulk flow are considered and all dissipative effects are integrated into a global damping factor λ. In contrast to most previous studies, which examine the interaction of infinitely many wave modes and truncate the system *a posteriori* [10, 11], we calculate the two-wave interaction directly allowing us to derive exact analytical solutions for the growth rates and stability onsets. The disadvantage of this method is that the MPR instability can sometimes arise from three (or more)-wave interactions [11, 17]. Accordingly, these cases are not represented in our model. A standard two-wave ansatz is given as follows:

$$[\boldsymbol{u}_i, p_i, \eta, \boldsymbol{j}_i] = \left(C\left[\widehat{\boldsymbol{u}}_i, \widehat{p}_i, \widehat{\eta}, \widehat{\boldsymbol{j}}_i\right] + C'\left[\widehat{\boldsymbol{u}}_i', \widehat{p}_i', \widehat{\eta}', \widehat{\boldsymbol{j}}_i'\right] + \left[\widetilde{\boldsymbol{u}}_i, \widetilde{p}_i, \widetilde{\eta}, \widetilde{\boldsymbol{j}}_i\right] \right) \mathrm{e}^{\left(\alpha - \bar{\lambda} + \mathrm{i}\bar{\omega}\right)t}. \tag{9}$$

In this notation, the tilded variables are small perturbations with respect to the hatted variables representing the basic two-wave state. The bare and the dashed($'$) hatted variables denote the variables corresponding to the two wave mode (m, n) and (m', n') accordingly. The eigenvalues C and C' are to be determined and α is the potentially destabilizing contribution to the growth rate that we seek to derive. We assume that both waves are magnetohydrodynamically coupled within their base state and therefore both oscillate with the same mean frequency. We represent the average natural frequency of the wave as $\bar{\omega} = (\omega + \omega')/2$ and attenuation with the mean damping rate as $\bar{\lambda} = (\lambda + \lambda')/2$. Similarly, the average frequency difference as $\delta\omega = (\omega - \omega')/2$ and decay rate difference as $\delta\lambda = (\lambda - \lambda')/2$. The wave pair is MPR unstable if and only if α is both a real number and exceeding the mean damping rate $\alpha > \bar{\lambda}$. The total growth rate α_{MPR} is then given by the difference $\alpha_{\mathrm{MPR}} = \alpha - \bar{\lambda}$ and stability onsets are defined by the condition $\alpha = \bar{\lambda}$. Considering the wave pair to be in a coupled motion $\omega = \omega' = \bar{\omega}$ is justified by the fact that the wave frequencies are shifted by Lorentz forces ($\bar{\omega} = \omega - \delta\omega$ and $\bar{\omega} = \omega' + \delta\omega$) in such a way that they coincide exactly at the point of marginal stability [12]. In the absence of any Lorentz forces, precisely the classic gravity-capillary wave modes given in Eq. (9) result as two independent solutions of our system. This rigorously justifies our approach, and our theory is capable of describing the complete transition from purely hydrodynamic waves passing through magnetohydrodynamically modified waves regimes up to the (linear) self-amplifying MPR wave.

To formulate the base state, we assign the hatted quantities with the classical gravity-capillary wave solutions that can be obtained by expressing the bulk velocities as gradients of scalar potentials $\widehat{\boldsymbol{u}}_i = \boldsymbol{\nabla}\widehat{\phi}_i$ and solving the Laplace equation $\boldsymbol{\nabla}^2\widehat{\phi}_i = 0$ together with the no outflow boundary condition [Eq. (6)] and the interface conditions given in Eqs. (8a) and (8b) [33]. The linear inviscid solutions for perturbation quantities of a singular wave mode (m, n) can be represented as follows:

$$\begin{bmatrix} \widehat{\boldsymbol{u}}_i \\ \widehat{p}_i \\ \widehat{\eta} \end{bmatrix} = \begin{bmatrix} \boldsymbol{\nabla}\widehat{\phi}_i \\ -\rho_i(\mathrm{i}\omega)\widehat{\phi}_i \\ \partial_z\widehat{\phi}_i/(\mathrm{i}\omega) \end{bmatrix}, \tag{10}$$

with the leading-order flow potentials given as

$$\begin{bmatrix} \widehat{\phi}_1 \\ \widehat{\phi}_2 \end{bmatrix} = A_{mn} \begin{bmatrix} \dfrac{\omega}{k}\dfrac{\cosh[k(z-h_1)]}{\sinh[kh_1]} \\ -\dfrac{\omega}{k}\dfrac{\cosh[k(z+h_2)]}{\sinh[kh_2]} \end{bmatrix} \cos\left[\frac{m\pi}{L_x}\left(x + \frac{L_x}{2}\right)\right] \cos\left[\frac{n\pi}{L_y}\left(y + \frac{L_y}{2}\right)\right] \tag{11}$$

along with the interface elevation function reading

$$\widehat{\eta} = \mathrm{i}\, A_{mn} \cos\left[\frac{m\pi}{L_x}\left(x + \frac{L_x}{2}\right)\right] \cos\left[\frac{n\pi}{L_y}\left(y + \frac{L_y}{2}\right)\right]. \tag{12}$$

Here, A_{mn} are mode-dependent wave amplitudes, k is a function of the wave numbers $(m, n) \in \mathbb{N}^0 \times \mathbb{N}^0$

$$k = \pi\sqrt{\frac{m^2}{L_x^2} + \frac{n^2}{L_y^2}} \tag{13}$$

and the natural frequencies of the gravity-capillary waves are obtained as

$$\omega = \pm\sqrt{\frac{(\rho_2 - \rho_1)gk + \gamma_{\mathrm{int}}k^3}{\rho_1 \coth[k\, h_1] + \rho_2 \coth[k\, h_2]}}. \tag{14}$$

It remains to calculate the leading-order current perturbations through Ohm's law $\widehat{\boldsymbol{j}}_i = -\sigma_i\boldsymbol{\nabla}\widehat{\varphi}_i$. The electric potentials must also fulfill the Laplace equation $\boldsymbol{\nabla}^2\widehat{\varphi} = 0$ with boundary conditions given in Eqs. (5a – 5d), (8c) and (8d). As the electrical boundary conditions at the insulating side walls are identical to the hydrodynamic outflow condition [Eq. (6)], the solutions of $\widehat{\varphi}_i$ result in a very similar form:

$$\begin{bmatrix} \widehat{\varphi}_1 \\ \widehat{\varphi}_2 \end{bmatrix} = J_0\, \Lambda\, \widehat{\eta} \begin{bmatrix} -\dfrac{\sinh[k(z-h_1)]}{\sigma_1 \cosh[kh_1]} \\ -\dfrac{\cosh[k(z+h_2)]}{\sigma_2 \sinh[kh_2]} \end{bmatrix}, \tag{15}$$

where Λ is

$$\Lambda = \frac{(\sigma_1^{-1} - \sigma_2^{-1})}{\sigma_1^{-1}\ \tanh[kh_1] + \sigma_2^{-1}\ \coth[kh_2]}, \tag{16}$$

and is called the conductivity-jump parameter. It does not appear in most MPR models because of the $\sigma_2 \gg \sigma_1$ approximation, however, this assumption is not always guaranteed in model MPR experiments [23]. It can be utilized to assess the minimum conductivity jump to trigger the MPR instability.

D. Calculation of the growth rate

All the preparations have now been made to calculate the growth rate α_{MPR}. By inserting the ansatz [Eq. (9)] in the governing Eqs. (2a) and (2b), and considering only the leading order terms, we can obtain the next order in the perturbation problem through the equations:

$$\begin{aligned} &C(\mathrm{i}\bar{\omega} - \bar{\lambda} + \alpha)\rho\widehat{\boldsymbol{u}} + C\boldsymbol{\nabla}\widehat{p} + C'(\mathrm{i}\bar{\omega} - \bar{\lambda} + \alpha)\rho\widehat{\boldsymbol{u}}' + C'\boldsymbol{\nabla}\widehat{p}' + (\mathrm{i}\bar{\omega} - \bar{\lambda})\rho\widetilde{\boldsymbol{u}} + \boldsymbol{\nabla}\widetilde{p} \\ &+(\mathrm{i}\bar{\omega} - \bar{\lambda})\rho\widetilde{\boldsymbol{u}}' + \boldsymbol{\nabla}\widetilde{p}' = C\left(\widehat{\boldsymbol{j}} \times B_z\boldsymbol{e}_z\right) + C'\left(\widehat{\boldsymbol{j}}' \times B_z\boldsymbol{e}_z\right), \end{aligned} \tag{17a}$$

$$\boldsymbol{\nabla} \cdot \widetilde{\boldsymbol{u}}_i = 0. \tag{17b}$$

To find an equation for α, we have to state a solvability condition. For that we apply the Fredholm alternative in an analogue manner to Herreman *et al.* [27]. All technical calculations are provided in the supplementary material in detail. The solvability condition results in an eigenvalue problem in form of a 2×2 interaction matrix that can be expressed in the following form:

$$\begin{bmatrix} \dfrac{I_{\eta\eta}}{2K_{\eta\eta}} + \mathrm{i}\delta\omega - \delta\lambda - \alpha & \dfrac{I_{\eta\eta'}}{2K_{\eta\eta}} \\ \dfrac{I_{\eta'\eta}}{2K_{\eta'\eta'}} & \dfrac{I_{\eta'\eta'}}{2K_{\eta'\eta'}} - \mathrm{i}\delta\omega + \delta\lambda - \alpha \end{bmatrix} \begin{bmatrix} C \\ C' \end{bmatrix} = 0, \tag{18}$$

with,

$$I_{\eta\eta} = \sum_{i=1,2} \int_{V_i} \widehat{\boldsymbol{u}}_i^* \cdot \left(\widehat{\boldsymbol{j}}_i \times B_z\boldsymbol{e}_z\right) = \sum_{i=1,2} \int_{V_i} \boldsymbol{\nabla}\widehat{\phi}_i^* \cdot \left(\sigma_i\boldsymbol{\nabla}\widehat{\varphi}_i \times B_z\boldsymbol{e}_z\right) = 0, \tag{19a}$$

$$I_{\eta\eta'} = \sum_{i=1,2} \int_{V_i} \widehat{\boldsymbol{u}}_i^* \cdot \left(\widehat{\boldsymbol{j}}_i' \times B_z\boldsymbol{e}_z\right) = \sum_{i=1,2} \int_{V_i} \boldsymbol{\nabla}\widehat{\phi}_i^* \cdot \left(\sigma_i\boldsymbol{\nabla}\widehat{\varphi}_i' \times B_z\boldsymbol{e}_z\right), \tag{19b}$$

$$I_{\eta'\eta} = \sum_{i=1,2} \int_{V_i} \widehat{\boldsymbol{u}}_i'^* \cdot \left(\widehat{\boldsymbol{j}}_i \times B_z\boldsymbol{e}_z\right) = \sum_{i=1,2} \int_{V_i} \boldsymbol{\nabla}\widehat{\phi}_i'^* \cdot \left(\sigma_i\boldsymbol{\nabla}\widehat{\varphi}_i \times B_z\boldsymbol{e}_z\right), \tag{19c}$$

$$I_{\eta'\eta'} = \sum_{i=1,2} \int_{V_i} \widehat{\boldsymbol{u}}_i'^* \cdot \left(\widehat{\boldsymbol{j}}_i' \times B_z\boldsymbol{e}_z\right) = \sum_{i=1,2} \int_{V_i} \boldsymbol{\nabla}\widehat{\phi}_i'^* \cdot \left(\sigma_i\boldsymbol{\nabla}\widehat{\varphi}_i' \times B_z\boldsymbol{e}_z\right) = 0. \tag{19d}$$

The terms given above are associated with the power that can be injected by Lorentz forces. Eqs. (19a) and (19d) describe the interaction of the wave modes η and η' with their self-induced Lorentz forces. However, these integrals disappear, which explains why standing waves cannot be destabilized. The terms $K_{\eta\eta}$ is associated with the potential (or equivalently kinetic) energy of the respective wave modes (m, n) and is given by

$$\begin{aligned} K_{\eta\eta} &= \int_{\mathcal{S}} \left(-\gamma_{\mathrm{int}}|\boldsymbol{\nabla}\widehat{\eta}|^2 + (\rho_2 - \rho_1)g|\widehat{\eta}|^2\right)\Big|_{z=0} \mathrm{d}S, \\ &= \xi_{m\times n}\, L_x L_y \left[\gamma_{\mathrm{int}}k^2 + (\rho_2 - \rho_1)g\right], \end{aligned} \tag{20}$$

Similarly, $K_{\eta\eta}$ is a function of (m', n') and k'. For the sake of simplicity, we introduced an auxiliary function ξ_r that is defined as follows

$$\xi_r = \begin{cases} 1/2, & r = 0 \\ 1/4, & r \neq 0 \end{cases}.$$

Finally, the secular equation of the matrix [Eq. (18)] yields the solutions for α

$$\alpha = \pm \sqrt{\frac{I_{\eta\eta'} I_{\eta'\eta}}{4 K_{\eta\eta} K_{\eta'\eta'}} + \left[\mathrm{i}\delta\omega - \delta\lambda\right]^2}. \tag{21}$$

By subtracting the damping rate $\bar{\lambda}$ and solving the integrals in Eqs. (19b) and (19c), we obtain the MPR growth rate, the central result of our study, that can be represented as follows

$$\alpha_{\mathrm{MPR}} = \alpha - \bar{\lambda} = \pm \sqrt{J^2 {B_z}^2 \Theta^2 \left(\bar{\omega}^2 - \delta\omega^2\right) \frac{\mathcal{T}\mathcal{T}'}{\mathcal{U}\mathcal{U}'} + \left[\mathrm{i}\delta\omega - \delta\lambda\right]^2} \; - \; \bar{\lambda}, \tag{22}$$

where

$$\Theta = \pm \sqrt{\delta_{m\times m'} \; \delta_{n\times n'}} \; \frac{((-1)^{m+m'} - 1)((-1)^{n+n'} - 1)(m^2 {n'}^2 - {m'}^2 n^2)}{(m^2 - {m'}^2)(n^2 - {n'}^2)}, \tag{23a}$$

$$\delta_r = \begin{cases} 1, & r = 0 \\ 2, & r > 0 \end{cases}. \tag{23b}$$

The term Θ here defines the wave mode selection function, which is nonzero only if $m + m'$ and $n + n'$ are both odd, and

$$\mathcal{U} = L_x L_y \Big(\gamma_{\mathrm{int}}\, k^2 + (\rho_2 - \rho_1) g\Big), \qquad \mathcal{U}' = L_x L_y \Big(\gamma_{\mathrm{int}}\, {k'}^2 + (\rho_2 - \rho_1) g\Big), \tag{23c}$$

are representations of the potential energy of the standing waves and

$$\mathcal{T} = \frac{\Lambda}{k(k^2 - k'^2)} \left(\frac{k}{\tanh[k' h_2]} - \frac{k'}{\tanh[k h_2]} + \frac{k' \; \mathrm{sech}[k' h_1]}{\sinh[k h_1]} - \frac{k'}{\tanh[k h_1]} + k \tanh[k' h_1] \right), \tag{23d}$$

$$\mathcal{T}' = \frac{\Lambda'}{k'({k'}^2 - k^2)} \left(\frac{k'}{\tanh[k h_2]} - \frac{k}{\tanh[k' h_2]} + \frac{k \; \mathrm{sech}[k h_1]}{\sinh[k' h_1]} - \frac{k}{\tanh[k' h_1]} + k' \tanh[k h_1] \right), \tag{23e}$$

root from the power injected by the Lorentz force and contain the complex non-shallow layer height dependencies of the cell currents redistributed by interface elevations. When $k' \to k$, these terms simplify to

$$\mathcal{T} = \mathcal{T}' = \frac{\Lambda}{4k^2 \; \sinh^2[k h_2]} \left(\frac{\sinh[k(h_1 + 2h_2)]}{\cosh[k h_1]} + 2k h_2 - \tanh[k h_1] \right). \tag{23f}$$

The perturbation analysis results in two distinct solutions for the growth rate α_{MPR}. The two solutions correspond to MPR waves that rotate either clockwise or anti-clockwise (when looking at the cell from above), or equivalently, to standing wave pairs $(\eta + \eta')$ with a positive or negative phase shift. One of the two wave solutions is always suppressed by the Lorentz force (when the induced Lorentz force $\boldsymbol{j} \times B_z \boldsymbol{e}_z$ opposes the rotating wave motion $\boldsymbol{u}$) and the other one is amplified in the unstable situations (when $\boldsymbol{j} \times B_z \boldsymbol{e}_z$ aligns with $\boldsymbol{u}$). In our configuration shown in Fig. 2, anti-clockwise rotating waves are destabilized.

From the growth rate Eq. (22) one can readily derive the stability onset that is defined by the equivalence of the real part of the square root in [Eq. (22)] with the average damping rate $\bar{\lambda}$. In order to find a preferably simple expression, we introduce a modified Sele parameter $\beta_{\gamma_{\mathrm{int}}}$, similar to Horstmann *et al.* [6], which takes into account interfacial tension in addition to gravity. In this way, the stability criterion can be represented compactly in a dimensionless form as follows

$$\beta_{\gamma_{\mathrm{int}}} > \beta_{\mathrm{crit}} = \frac{L_x^2 L_y^2}{h_1 h_2 \Theta} \sqrt{\frac{\bar{\lambda}^2 - \left[\mathrm{i}\delta\omega - \delta\lambda\right]^2}{(\bar{\omega}^2 - \delta\omega^2)\mathcal{T}\mathcal{T}'}}, \quad \text{where } \beta_{\gamma_{\mathrm{int}}} = \frac{I_0 B_z}{\left[(\rho_2 - \rho_1) g + \gamma_{\mathrm{int}} \bar{k}^2\right] h_1 h_2}. \tag{24}$$

In order to analyse the stability of large-scale aluminum reduction cells, the criterion can be further simplified. We reformulate Eq. (24) in the shallow-water limit and neglect interfacial tension. Then the interfacial waves become non-dispersive ($\omega = ck$, $\omega' = ck'$), with

$$c = \sqrt{\frac{(\rho_2 - \rho_1)g}{\rho_1/h_1 + \rho_2/h_2}}. \tag{25}$$

The product $\mathcal{T}\mathcal{T}'$ simplifies as

$$\mathcal{T}\mathcal{T}' = \frac{1}{k^4 k'^4 h_1^2 h_2^2}, \tag{26}$$

and therewith Eq. (24) reduces to

$$\beta_{\text{crit}} = \frac{L_x^2 L_y^2}{c\,\Theta}\sqrt{k^3 k'^3 \left(\bar{\lambda}^2 - \left[\mathrm{i}\delta\omega - \delta\lambda\right]^2\right)}. \tag{27}$$

This stability threshold criterion coincides with the solution of linear MPR instability presented in Politis and Priede [28], see Sec. III C. As a final simplification, we shall take a look at the academic case of square cells, which is physically relevant for some MPR model experiments [19, 21, 22, 24]. For a square cell we have $L_x{=}L_y{=}L$, $\lambda{=}\lambda'{=}\bar{\lambda}$ and the parameters further reduce to $\Theta = 4$, $k = \pi/L$, $\delta\omega = 0$, $\delta\lambda = 0$, yielding

$$\beta_{\text{crit}} = \frac{\pi^3 L}{4\,c}\,\lambda. \tag{28}$$

As expected, the critical Sele parameter in this case is not dictated by the difference in natural frequencies anymore but solely by the damping parameter λ, as is true for all infinitely many inherently unstable aspect ratios, refer to Sec. III C, Eq. (45).

E. The viscous damping rate

Up to this point, we have been treating λ as a purely phenomenological damping parameter. To evaluate the stability of industrial ARCs, in which turbulent flows usually prevail, λ must be estimated or determined experimentally. In small-scale models, where flows remain largely laminar (at least in the boundaries or around the point of marginal instability), viscous damping rates can be calculated analytically. This is a key ingredient for our model to serve as a benchmark. We start with the viscous damping rate. In the leading order, magnetohydrodynamics effects can be neglected (no consideration of Hartmann layers) and it remains to calculate the total dissipation produced in the Stokes boundary layers, which are created solely by wave motions. We follow the approach by Keulegan [29], who has calculated damping rates of free-surface gravity waves in rectangular containers. We extend his formulation to two liquid layers, which is not straightforward as viscous boundary layers are formed above and below the interface in two-liquid systems that do not develop on free surfaces and must be calculated separately. In absence of MHD effects, the linear governing Eqs. (2a) and (2b) reduce to

$$\left.\begin{aligned} \rho_i \partial_t \boldsymbol{u}_i + \boldsymbol{\nabla} p_i &= \rho_i \nu_i \nabla^2 \boldsymbol{u}_i \\ \boldsymbol{\nabla}\cdot\boldsymbol{u}_i &= 0 \end{aligned}\right\}. \tag{29}$$

For the upper and lower layer ($i = 1, 2$), the no slip condition at the side wall (represented as rigid boundary Σ_i) is

$$\boldsymbol{u}_i\big|_{\Sigma_i} = 0, \tag{30a}$$

and the tangential velocities (denoted by $\perp$) must be equal at the interface:

$$\boldsymbol{u}_{1,\perp}\big|_{\mathcal{S}} = \boldsymbol{u}_{2,\perp}\big|_{\mathcal{S}}. \tag{30b}$$

The dynamic boundary condition [Eq. (8b)] under the consideration of viscous effects now changes to

$$p_2|_{\mathcal{S}} - p_1|_{\mathcal{S}} = (\rho_2 - \rho_1)g\eta - \gamma_{\text{int}}\nabla^2\eta + 2(\rho_2\nu_2\partial_z u_{2,z} - \rho_1\nu_1\partial_z u_{2,z}), \tag{30c}$$

while the continuity of the tangential stresses are

$$\rho_1\nu_1\Big(\partial_z u_{1,x}+\partial_x u_{1,z}\Big)\Big|_{\mathcal{S}}=\rho_2\nu_2\Big(\partial_z u_{2,x}+\partial_x u_{2,z}\Big)\Big|_{\mathcal{S}}, \tag{30d}$$

$$\rho_1\nu_1\Big(\partial_z u_{1,y}+\partial_y u_{1,z}\Big)\Big|_{\mathcal{S}}=\rho_2\nu_2\Big(\partial_z u_{2,y}+\partial_y u_{2,z}\Big)\Big|_{\mathcal{S}}. \tag{30e}$$

In a stratified two-layer system, consider an interfacial gravity wave mode of (m,n) of amplitude A_{mn} oscillating at angular frequency ω undergoes viscous decay over time at a rate of λ. The fluid velocity taking into account the Stokes boundary layer corrections can be formulated as

$$\boldsymbol{u}_i = C\Big[\widehat{\boldsymbol{u}}_i+\widehat{\bar{\boldsymbol{u}}}_i\Big]e^{(\mathrm{i}\omega-\lambda)t}. \tag{31}$$

where the $\widehat{\boldsymbol{u}}_i$ are the potential flow velocities and the barred term $\widehat{\bar{\boldsymbol{u}}}_i$ are the viscous corrections to be calculated. We introduce wall-normal coordinates ζ that tend to decay the viscous boundary layer corrections to zero as they traverse inwards in the bulk. We define ζ as follows:

$$\zeta=\begin{cases} z-h_1, & \text{near } z=h_1\\ \dfrac{L_x}{2}-x, & \text{near } x=L_x/2\\ \dfrac{L_y}{2}-y, & \text{near } y=L_y/2\\ z, & \text{near } z=0\end{cases}, \qquad \zeta=\begin{cases} z+h_2, & \text{near } z=-h_2\\ x+\dfrac{L_x}{2}, & \text{near } x=-L_x/2\\ y+\dfrac{L_y}{2}, & \text{near } y=-L_y/2\\ -z, & \text{near } z=0\end{cases}. \tag{32}$$

The effect of the diffusive boundary layer tends to diminish in the bulk region. Inserting the ansatz [Eq. (31)] into Eq. (29), we obtain $\widehat{\bar{\boldsymbol{u}}}_i\cdot\boldsymbol{n}_i=0$ and $\widehat{\bar{p}}_i=0$; the wall-normal boundary layer correction for velocity and leading-order boundary layer correction for pressure, both equal to zero. We compute the Stokes boundary layer at the domain walls (with no-slip condition) as

$$\mathrm{i}\omega\widehat{\bar{\boldsymbol{u}}}_{i,\perp}\approx\nu_i\partial^2_{\zeta\zeta}\widehat{\bar{\boldsymbol{u}}}_{i,\perp}. \tag{33}$$

The mode-dependent velocity at the solid walls are

$$\widehat{\bar{\boldsymbol{u}}}_{i,\perp}=-\widehat{\boldsymbol{u}}_{i,\perp}|_{\Sigma_i}\mathrm{e}^{-\Gamma\sqrt{|\omega|/\nu_i}\zeta-\lambda t}, \qquad \Big\{\Gamma=\big(1+\mathrm{i}\omega\big)/\sqrt{2}. \tag{34}$$

From Eq. (34) we can obtain the characteristic length of the viscous boundary layer of each medium as $\sqrt{\nu_i/|\omega|}$. At the interface, from Eqs. (30d) and (30e) we deduce

$$\widehat{\boldsymbol{u}}_{1,\perp}+\widehat{\bar{\boldsymbol{u}}}_{1,\perp}|_{\mathcal{S}}\approx\widehat{\boldsymbol{u}}_{2,\perp}+\widehat{\bar{\boldsymbol{u}}}_{2,\perp}|_{\mathcal{S}} \qquad \begin{cases}\rho_1\nu_1\partial_z\widehat{\bar{u}}_{1,x}|_{\mathcal{S}}\approx\rho_2\nu_2\partial_z\widehat{\bar{u}}_{2,x}|_{\mathcal{S}}\\ \rho_1\nu_1\partial_z\widehat{\bar{u}}_{1,y}|_{\mathcal{S}}\approx\rho_2\nu_2\partial_z\widehat{\bar{u}}_{2,y}|_{\mathcal{S}}\end{cases}, \tag{35}$$

$$\widehat{\bar{\boldsymbol{u}}}_{1,\perp}\Big|_{m,n}=\frac{\Lambda_{\mathrm{v}}}{\rho_1\sqrt{\nu_1}}\,\frac{\omega}{k}\begin{bmatrix}\frac{m\pi}{L_x}\ \sin\left[\frac{m\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right]\cos\left[\frac{n\pi}{L_y}\left(y+\frac{L_y}{2}\right)\right]\boldsymbol{e}_x\\ +\\ \frac{n\pi}{L_y}\ \cos\left[\frac{m\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right]\sin\left[\frac{n\pi}{L_y}\left(y+\frac{L_y}{2}\right)\right]\boldsymbol{e}_y\end{bmatrix}\mathrm{e}^{-\Gamma\sqrt{\frac{|\omega|}{\nu_1}}z-\lambda t}, \tag{36}$$

$$\widehat{\bar{\boldsymbol{u}}}_{2,\perp}\Big|_{m,n}=\frac{-\Lambda_{\mathrm{v}}}{\rho_2\sqrt{\nu_2}}\,\frac{\omega}{k}\begin{bmatrix}\frac{m\pi}{L_x}\ \sin\left[\frac{m\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right]\cos\left[\frac{n\pi}{L_y}\left(y+\frac{L_y}{2}\right)\right]\boldsymbol{e}_x\\ +\\ \frac{n\pi}{L_y}\ \cos\left[\frac{m\pi}{L_x}\left(x+\frac{L_x}{2}\right)\right]\sin\left[\frac{n\pi}{L_y}\left(y+\frac{L_y}{2}\right)\right]\boldsymbol{e}_y\end{bmatrix}\mathrm{e}^{\Gamma\sqrt{\frac{|\omega|}{\nu_2}}z-\lambda t}, \tag{37}$$

with, $\quad \Lambda_{\mathrm{v}}=\dfrac{(\coth[kh_1]+\coth[kh_2])}{1/\big(\rho_1\sqrt{\nu_1}\big)+1/\big(\rho_2\sqrt{\nu_2}\big)}.$

The energy balance [Eq. (38)] for viscous dissipation can be equated to the rate of change of kinetic and potential energy. It is given as

$$\frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\left(\sum_{i=1,2}\frac{1}{2}\int_{V_i}\rho_i|\boldsymbol{u}_i|^2\mathrm{d}V]\right)}_{E_{\mathrm{kin}}}+\frac{\mathrm{d}}{\mathrm{d}t}\underbrace{\left(\int_S\left[\frac{1}{2}(\rho_2-\rho_1)g\eta^2+\gamma_{\mathrm{int}}\sqrt{1+|\nabla\eta|^2}\right]\mathrm{d}S\right.}_{E_{\mathrm{pot}}}$$
$$=-\underbrace{\sum_{i=1,2}2\rho_i\nu_i\int_{V_i}\boldsymbol{\varepsilon}_i:\boldsymbol{\varepsilon}_i\,\mathrm{d}V}_{\mathcal{D}}. \tag{38}$$

We adapt the dissipation function $\mathcal{D}$ from Lamb [33], in which the strain tensor is denoted by $\boldsymbol{\varepsilon}_i$. The solution to the balance Eq. (38) provides the damping rates due to the side wall (λ_{wall}), interface (λ_{int}), and irrotational viscous stresses (λ_{irr}). The total damping rate for a wave of mode (m,n) is composed as

$$\lambda=\lambda_{\mathrm{wall}}+\lambda_{\mathrm{int}}+\lambda_{\mathrm{irr}}, \tag{39}$$

and is assumed to have a much larger time scale than the natural frequency of the rotating wave. All calculations are given in detail in the supplementary material. The solutions of λ_{wall}, λ_{int} and λ_{irr} are

$$\lambda_{\mathrm{wall}}(m,n)=\sum_{i=1,2}\frac{\frac{1}{\sqrt{2}}\frac{\rho_i\sqrt{\omega\nu_i}}{kL_xL_y}}{\left[\frac{\rho_1}{\tanh[kh_1]}+\frac{\rho_2}{\tanh[kh_2]}\right]}\left(\left[\frac{L_xL_y\,k^2}{2\sinh^2[kh_i]}+\frac{\delta_m h_i(n^2\pi^2-k^2L_y^2)}{2L_y\,\sinh^2[kh_i]}\right.\right.$$
$$\left.\left.+\frac{\delta_m(n^2\pi^2+k^2L_y^2)}{2L_y\,k\,\tanh[kh_i]}+\frac{\delta_n h_i(m^2\pi^2-k^2L_x^2)}{2L_x\,\sinh^2[kh_i]}+\frac{\delta_n(m^2\pi^2+k^2L_x^2)}{2L_x\,k\,\tanh[kh_i]}\right]\right), \tag{40}$$

$$\lambda_{\mathrm{int}}(m,n)=\frac{k\sqrt{\omega}}{2\sqrt{2}}\frac{\left(\coth[kh_1]+\coth[kh_2]\right)^2}{\left(\frac{1}{\rho_1\sqrt{\nu_1}}+\frac{1}{\rho_2\sqrt{\nu_2}}\right)\left[\frac{\rho_1}{\tanh[kh_1]}+\frac{\rho_2}{\tanh[kh_2]}\right]}, \tag{41}$$

$$\lambda_{\mathrm{irr}}(m,n)=\frac{2k^2\left(\frac{1}{\rho_1\sqrt{\nu_1}}+\frac{1}{\rho_2\sqrt{\nu_2}}\right)^{-1}}{\rho_1\coth[kh_1]+\rho_2\coth[kh_2]}\left(\left[\frac{\rho_2\sqrt{\nu_1\nu_2}+\rho_1\nu_1}{\rho_2\sqrt{\nu_2}\tanh[kh_1]}+\frac{\rho_1\sqrt{\nu_1\nu_2}+\rho_2\nu_2}{\rho_1\sqrt{\nu_1}\tanh[kh_2]}\right.\right.$$
$$\left.\left.-\left(\coth[kh_1]+\coth[kh_2]\right)\left(\sqrt{\nu_1}+\sqrt{\nu_2}\right)\right]\right). \tag{42}$$

The Eq. (40) is the mode-dependent damping rate originating from the bounding walls. The first term of the five-part sum corresponds to the damping at the top and bottom walls. The remaining terms stem from viscous dissipation in the lateral walls (see Eq. (23b) for δ_m and δ_n functions). In model ARCs, which have smaller lateral and longitudinal surface areas than the upper and lower ones, dissipation at the top and bottom surfaces tend to outweigh the contribution from the sidewalls. This is contrary to LMBs which, in general, have a larger sidewall surface area than the upper/lower surface area. The damping at the interface [Eq. (41)] is the same as in cylindrical cells [27]. Classically, when considering free-surface waves, the irrotational term in Eq. (42) considers the dissipation in the irrotational bulk flow. In our formulation, it also involves some leading-order contributions at the interface, which do not appear in gas-liquid free surface systems. It should be noted that the presented contributions are only applicable when we assume no contact angle hysteresis and no-slip boundary condition at the bounding walls.

In the single layer limit $\rho_1=0$, interface damping λ_{int} reduces to zero. Considering unidirectional wave modes $(m,0)$, Eqs. (40) and (42) simplify to

$$\lambda=\sqrt{\frac{\omega\nu_2}{2}}\frac{1}{L_y}\left[1+\frac{L_y}{L_x}\left(1+m\pi\frac{L_x-2h_2}{L_x\sinh[2m\pi h_2/L_x]}\right)\right]+2\nu_2k^2. \tag{43}$$

This single-layer wave limit is fully equivalent to the well-known free-surface damping rate derived by Keulegan [29] as presented in [34, chapter 6].

TABLE I. Cell parameters and material properties from the different studies, for comparison with the theoretical results.

Parameters from La Rocca *et al.* [30]		
Dimensions of cell ($L_x = L_y$)	$0.5\,\mathrm{m}$	
	Ω_1	Ω_2
Density (ρ_1, ρ_2)	$850\,\mathrm{kg\,m^{-3}}$	$1000\,\mathrm{kg\,m^{-3}}$
Kinematic viscosity (ν_1, ν_2)	$3.6 \times 10^{-5}\,\mathrm{m^2\,s^{-1}}$	$1.0 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$
Layer thickness (h_1, h_2)	$0.06\,\mathrm{m}$	$0.08\,\mathrm{m}$
Parameters from Politis and Priede [28]		
Length of cell (L_x)	$6.325\,\mathrm{m}$	
Width of cell (L_y)	$6.325, \ldots, 2.108$ m	
	Cryolite (Ω_1)	Aluminum (Ω_2)
Density (ρ_1, ρ_2)	$2130\,\mathrm{kg\,m^{-3}}$	$2330\,\mathrm{kg\,m^{-3}}$
Kinematic viscosity (ν_1, ν_2)	$4.4 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$	$8.8 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$
Electrical conductivity (σ_1, σ_2)	$210\,\mathrm{S\,m^{-1}}$	$3.3 \times 10^{6}\,\mathrm{S\,m^{-1}}$
Layer thickness (h_1, h_2)	$0.05\,\mathrm{m}$	$0.25\,\mathrm{m}$
Parameters from Borisov *et al.* [21]		
Cross-section area of cell (S)	$0.002\,\mathrm{m^2}$	
	Nitric acid (alky) (Ω_1)	Nitric acid (aq) (Ω_2)
Density (ρ_1, ρ_2) at 35°C	$859\,\mathrm{kg\,m^{-3}}$	$1050\,\mathrm{kg\,m^{-3}}$
at 45°C	$847\,\mathrm{kg\,m^{-3}}$	$1046\,\mathrm{kg\,m^{-3}}$
at 55°C	$834\,\mathrm{kg\,m^{-3}}$	$1041\,\mathrm{kg\,m^{-3}}$
Kinematic viscosity (ν_1, ν_2) at 35°C	$2.9 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$	$7.4 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$
at 45°C	$2.2 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$	$6.1 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$
at 55°C	$1.8 \times 10^{-6}\,\mathrm{m^2\,s^{-1}}$	$5.2 \times 10^{-7}\,\mathrm{m^2\,s^{-1}}$
Electrical conductivity (σ_1, σ_2)	$0.67\,\mathrm{S\,m^{-1}}$	$47.5\,\mathrm{S\,m^{-1}}$
Layer thickness ($h_1 = h_2$)	$0.014\,\mathrm{m}$	

III. COMPARISON AND VALIDATION

This chapter is dedicated to the validation of our model, which essentially consists of the two-wave MPR growth rate [Eq. (22)] and corresponding instability thresholds [Eq. (24)] combined with the independently derived viscous [Eq. (39)] damping rate. To this end we compare several predictions with different previous studies, whereby we also want to scrutinize some interesting aspects of the MPR physics deducible from our model.

First, in Sec. III A, we compare the analytical damping rate with numerical solver ARCHER, where the interface is modeled and tracked using a level set method coupled with volume of fluid (VOF) equations in Cartesian grids [35, 36]. In Sec. III B, we compare the viscous damping rate with the two-layer sloshing experiments and simulations by La Rocca *et al.* [30]. In Sec. III C, we show that our model is equivalent to the most recent MPR theory by Politis and Priede [28] in the limits of shallow-water gravity waves ($\gamma_{\mathrm{int}} = 0$) with constant damping and $\sigma_2 \gg \sigma_1$. Finally, in Sec. III D, our predictions for critical Sele parameters are compared with experimental measurements obtained in the two-layer MPR experiment undertaken by Borisov *et al.* [21] at several different aspect ratios. Table I summarizes all of the physical parameters and material properties taken from the respective studies that are required to reproduce all the results presented in this section.

A. Numerical simulations on the viscous damping rate

We have already shown that the viscous damping rate λ agrees with the established linear damping rate in Eq. (43) derived by Keulegan [29] within its single-layer limit describing free-surface waves. However, interfacial waves are physically different because the leading-order viscous boundary layers form above and below the interface (the viscous boundary layer below a free surface is of higher order [37]), and Lamb's dissipation integral must be treated differently. However, the literature on two-layer interfacial wave damping is unfortunately rather limited. Therefore, we cross-validated the analytical results from Eqs. (40–42) with DNS using ARCHER. For a two-layer (cryolite–aluminium) system considering geometries with smaller lateral surface areas and equal layer heights, we initialized a standing interfacial wave profile of mode (m, n) with a small amplitude of $L_z/500$ depending on the aspect ratio and let it decay over time. Three different cell aspect ratios (1, 2 and 3) with varying cell surface areas were simulated and

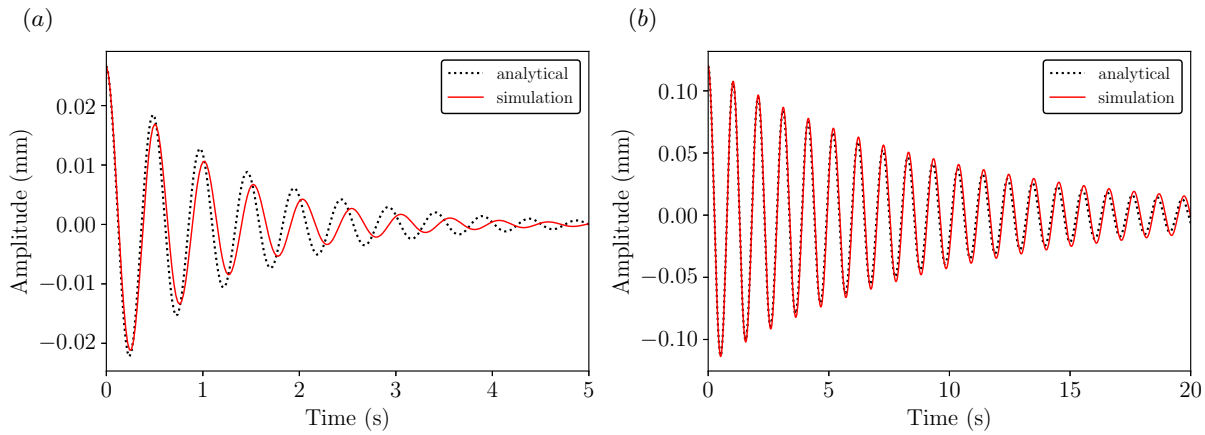

FIG. 3. The theoretical and simulated wave amplitude decay for mode $(4, 1)$ in a cell of length (a) $L_x = 4$ cm and (b) $L_x = 18$ cm and aspect ratio $L_x/L_y = 3$. The initial amplitude specified as $L_z/500$ m. The two-layer viscous damping equation succeeds in capturing the wave frequency and decay rate in the larger cell with a larger bulk flow domain.

compared with the theoretical values, each with a fixed length L_x and layer height equal to $h_1, h_2 = L_x/8$. The total surface areas were kept relatively small in order to reduce the computational cost. Due to the lack of sufficient models available to track the contact line motion of the interface, we were limited to an interfacial contact angle of 90° and therefore, had to neglect the capillary effects on the lateral wall. The value of the interfacial tension was also set to zero. Fig. 3 shows the simulated wave decay profile in comparison with the theoretical profile, for a particular case of mode $(4, 1)$ in two different cell geometries, $L_x = 4$ cm and $L_x = 18$ cm. The damping rate of the waves were obtained from fitting an exponential curve on the envelope of the respective numerical curve. A detailed explanation regarding the post processing of the decay rates is provided in Appendix A. The damping rates obtained for all wave modes pertaining to geometries $L_x = 4$ and $L_x = 18$ cm, are presented in Fig. 4. For the latter case ($L_x = 18$ cm), we found good agreement between the simulated and theoretical values of the damping rates, with a net deviation of less than 10%. This is the relevant case as as most mid-sized MPR setups have dimensions around or larger than $L_x = 18$ cm [19, 24]. The values of the damping rate obtained from simulating smaller cell geometries with $L_x = 4$ cm showed a larger disagreement with the theory, with an average error of 22%. Deviations in the wave frequency (see Fig. 3(a)) contributed largely to disagreement in the damping rates. The poor agreement in the wave frequency and damping rates in the smaller cells can be attributed to the viscous sub-layer $\sqrt{\nu_i/\omega}$ having the same scale as the inviscid bulk flow. In these cases, our boundary layer correction approach that relies on inviscid irrotational bulk solutions begins to lose its applicability [38]. The experiment by Borisov *et al.* [21] uses cells of similar dimensions but with double the layer height, where our approach is still a good approximation [Eq. (34)]. The comparisons of the damping rates for other cell geometries $L_x = 6, 12$ cm are provided in Table III, Appendix A.

B. Comparison with La Rocca *et al.* [30]

La Rocca *et al.* [30] conducted interfacial sloshing experiments in an enclosed square tank filled with two stably stratified liquid layers. The authors put one focus on dissipative effects and carried out direct decay rate measurements of different previously-excited interfacial wave modes. A theoretical model for viscous dissipation was also developed, without, however, providing an analytical expression for the damping rates. In Fig. 5 we compare Eq. (39) with La Rocca's results limited to cases where the authors provide both numerical and experimental data. Fig. 5(a) shows the non-dimensionalized viscous damping rate as a function of the non-dimensionalized kinematic viscosity for three different wave modes $(m, 0)$, $m = 1, 2, 3$. Our linear damping theory slightly overshoots La Rocca's predictions but agrees even better with the three experimental measurements carried out at $\nu_1/(gL_y^3)^{1/2} = 3.25 \times 10^{-5}$ and $\rho_1/\rho_2 = 0.85$.

The applicability of La Rocca's model, however is also limited, as is evident from Fig. 5(b), showing the damping rate as a function of the density ratio ρ_1/ρ_2. This figure portrays the transition from free-surface to interfacial wave damping. In the limit of quasi free surfaces $\rho_1/\rho_2 \lesssim 0.1$ we find, as expected, very good agreement with our model. Their damping rate estimates tend to diverge at $\rho_1 = \rho_2$, which is physically incorrect as the wave's natural frequencies become infinitely small in the limit $\rho_1 \to \rho_2$ and, correspondingly, the waves must then attenuate infinitely slowly. Since we have chosen exactly the same ansatz as Herreman *et al.* [27], who derived λ for cylindrical geometries, and

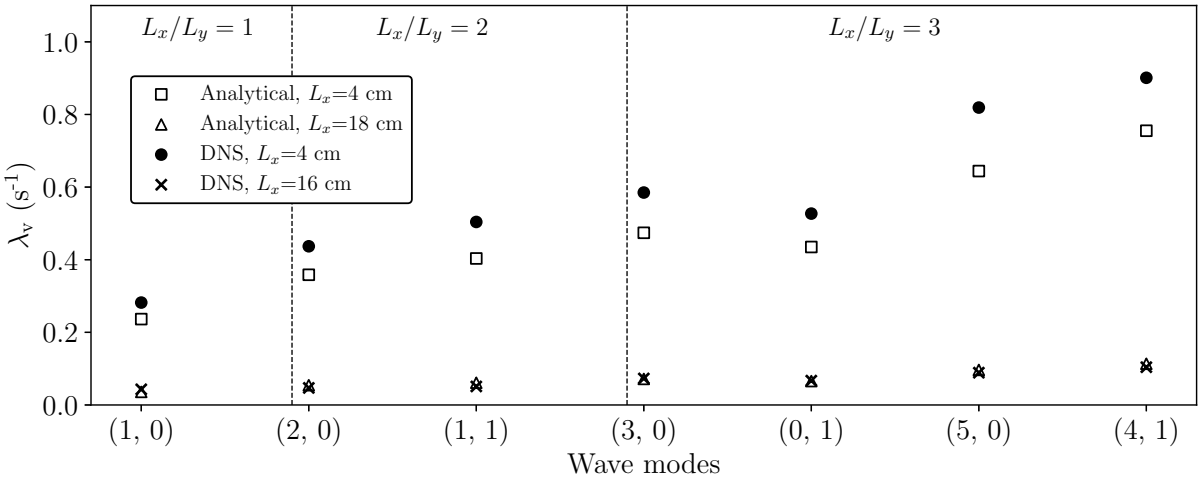

FIG. 4. The comparison of the viscous damping rates for a two-layer cryolite-aluminum system in a rectangular geometry with the analytical viscous decay rates. The respective dimensions of these systems are: length $L_x = 4$ cm and layer heights $h_1, h_2 = 0.5$ cm and length $L_x = 18$ cm and layer heights $h_1, h_2 = 2.25$ cm. The simulations were conducted in the most refined meshes obtained from grid dependency tests (see Appendix A).

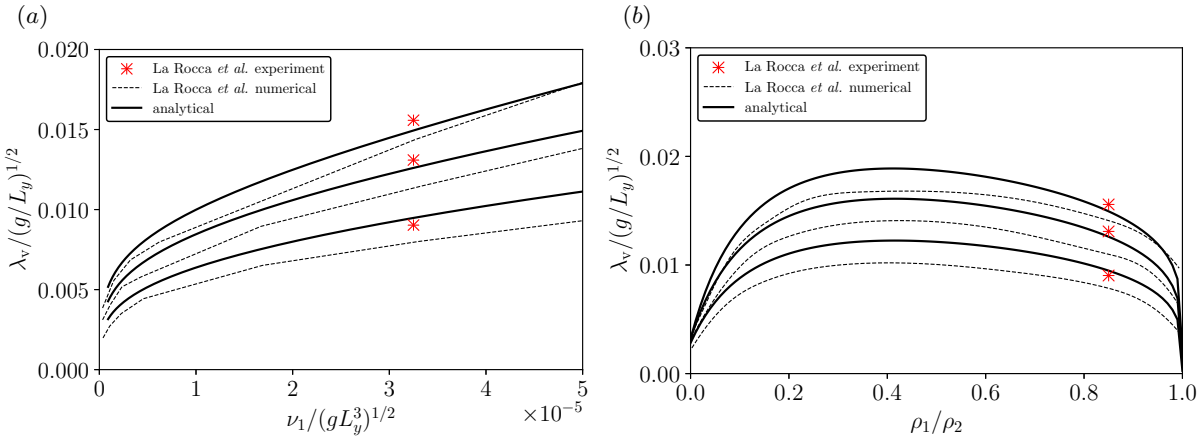

FIG. 5. (a) Non-dimensionalized viscous damping rate as a function of the non-dimensionalized kinematic viscosity for three different unidirectional wave modes $(m, 0)$, $m = 1, 2, 3$ at a constant density ratio of $\rho_1/\rho_2 = 0.85$. (b) non-dimensionalized damping rate as a function of the density ratio ρ_1/ρ_2 at a constant viscosity ratio ($\nu_1/\nu_2 \approx 36$) for the same wave modes. Additional measurement points are included for the first three modes $m = 1, 2, 3$.

the decay rates have been successfully validated against measurements by Horstmann *et al.* [39], and had given good agreement with DNS simulations, we are convinced that our treatment of viscous damping is reliable given the applied approximations, which also involves the Keulegan limit [Eq. 43].

C. Comparison with Politis and Priede [28]

We will show that the linear MPR growth rate and stability onset predictions by Politis and Priede [28] are fully contained in our model as a limiting case for $h_1, h_2 \ll L_x, L_y$, $\sigma_1 \ll \sigma_2$, $\gamma_{\text{int}} = 0$ and constant damping coefficients. First, as a special case, we consider the growths of dissipationless MPR waves in inherently unstable cell configurations, for which Politis and Priede [28] derived a compact analytical expression. A cell is called inherently unstable whenever there exist two wave modes $(m, n) + (m', n')$, such that $(m + n) \wedge (m' + n') = \mathbb{N}_{\text{odd}}$ and fulfilling $\delta\omega = 0$. In shallow geometries, gravity waves are non-dispersive ($\omega = ck$, $\omega' = ck'$) and the frequency equality $\delta\omega = 0$ reduces to a wave

number equality $k = k'$, imposing the following constraint on the lateral aspect ratio:

$$k = \pi\sqrt{\frac{m^2}{L_x^2} + \frac{n^2}{L_y^2}} = \pi\sqrt{\frac{m'^2}{L_x^2} + \frac{n'^2}{L_y^2}} = k' \implies \frac{L_x^2}{L_y^2} = \frac{m'^2 - m^2}{n^2 - n'^2} = \frac{(m'-m)(m'+m)}{(n-n')(n+n')}. \tag{44}$$

This condition allows us to easily understand why the unstable aspect ratios form an absolutely discontinuous dense set of points. For a simple explanation, let us first consider the wave mode function Θ [Eq. (23a)], which can only be non-zero if both wave mode pairs (m, m') and (n, n') are of opposite parity. Wave pairs whose wave modes (m, m') or (n, n') are both even or odd can therefore never be destabilized. From number theory, all sums and differences of numbers with opposite parity yield odd numbers $m \pm m' =$ odd and $n \pm n' =$ odd. Likewise, products of odd numbers are again always odd, which is why we can assign

$$\frac{(m'-m)(m'+m)}{(n-n')(n+n')} = \frac{M}{N}, \tag{45}$$

where M and N can be any two odd numbers. The fractality of the unstable aspect ratios is therefore, occurring from the constraint $L_x/L_y = \sqrt{M/N}$ [28]. From Eq. (45) we can deduce the most unstable wave pair for all cells fulfilling $L_x/L_y = \sqrt{M/N}$ in the inviscid limit:

$$m = \frac{M-1}{2}, \quad m' = \frac{M+1}{2}, \qquad n = \frac{N+1}{2}, \quad n' = \frac{N-1}{2}. \tag{46}$$

By inserting the most unstable wave numbers [Eq. (46)] into the growth rate solution in Eq. (22) and applying all the mentioned limits we obtain an expression for α_{MPR} as a function of the fractal numbers

$$\alpha_{\mathrm{MPR}} = \pm\,\beta_{\mathrm{S}}\,\frac{c}{\sqrt{L_x L_y}}\,\frac{8}{\pi^3}\sqrt{\frac{\delta_{M-1}\delta_{N-1}}{(M+N)(MN+1)\sqrt{MN}}}. \tag{47}$$

The above Eq. (47) (see Eq. (23b) for $\delta_{M-1}, \delta_{N-1}$) is the dimensional form of the growth rate solution [Eq. (3.28)] of Politis and Priede [28], which verifies the equivalence of both models. It remains to validate our predictions of stability onsets, which Politis and Priede [28] have presented in the form of critical Sele parameters β_{crit} and studied their dependence on the aspect ratio. For this purpose, the authors also modeled damping by introducing a phenomenological friction coefficient γ. However, the resulting eigenvalue problem was only solved numerically using the Galerkin method, and for this reason we compare the onset solutions graphically in the following. Although we have derived an explicit analytical solution for β_{crit}, Eq. (24), it is not evident *a priori* which mode pair is most unstable and thus defines β_{crit}. Outside the shallow-water approximation, the most unstable mode pair cannot be determined analytically. Our simple strategy for solving this problem is to calculate β_{crit} for all possible combinations of wave pairs and then to assign β_{crit} to the particular wave pair that yields the lowest onset value. The smallest possible critical Sele parameter defines the global instability onset. However, this iterative method is not a major limitation for our model because large-scale wave modes are always most unstable in practice, such that only wave numbers up to about $m, n \lesssim 10$ must be examined. This is because damping increases markedly with increasing wave numbers and also because small-scale wave motions are gradually suppressed by interfacial tension ($\beta_{\gamma_{\mathrm{int}}}$ in Eq. (24) diminishes with increasing k).

Fig. 6(a) shows iteratively calculated onset values β_{crit} as a function of the squared lateral aspect ratio $(L_x/L_y)^2$ compared with numerical solutions by Politis and Priede [28] for different friction coefficients γ. All geometrical parameters and material properties of the considered cell are given in Table I. We find a quasi-perfect agreement between both models for any given friction coefficient. The pattern of the green fractal curve ($\gamma = 0$) is determined solely by the chosen resolution. All other curves can be unambiguously determined; the non-differentiable inflection points indicate the transitions between different most unstable wave pairs. Fig. 6(b) provides the analytical prediction from Eq. (24) taking into account wave mode-dependent viscous damping from Eq. (39), highlighting all the unstable wave modes between the inflection points. Extending the graph until the aspect ratio square of 21 shows us why most industrial ARCs operate at $(L_x/L_y)^2 = 17$–18, due to the presence of a larger stability buffer [40]. Using Fig. 6(b), it is possible to explain the small deviation emerging between our respective theories at $\gamma = 0.2$ for $(L_x/L_y)^2 > 8.45$. The curve from Politis and Priede [28] shows here β_{crit} for the mode pair $(5, 0) + (4, 1)$. However, our solution predicts that the pair $(3, 0) + (0, 1)$ becomes unstable first, and accordingly our calculated curve is slightly lower. Apart from this point, however, the most unstable wave pair is always correctly predicted by the wave number tuples [Eq. 46]. It is important to note that the absolute values of the critical beta are not very significant in large cells. However,

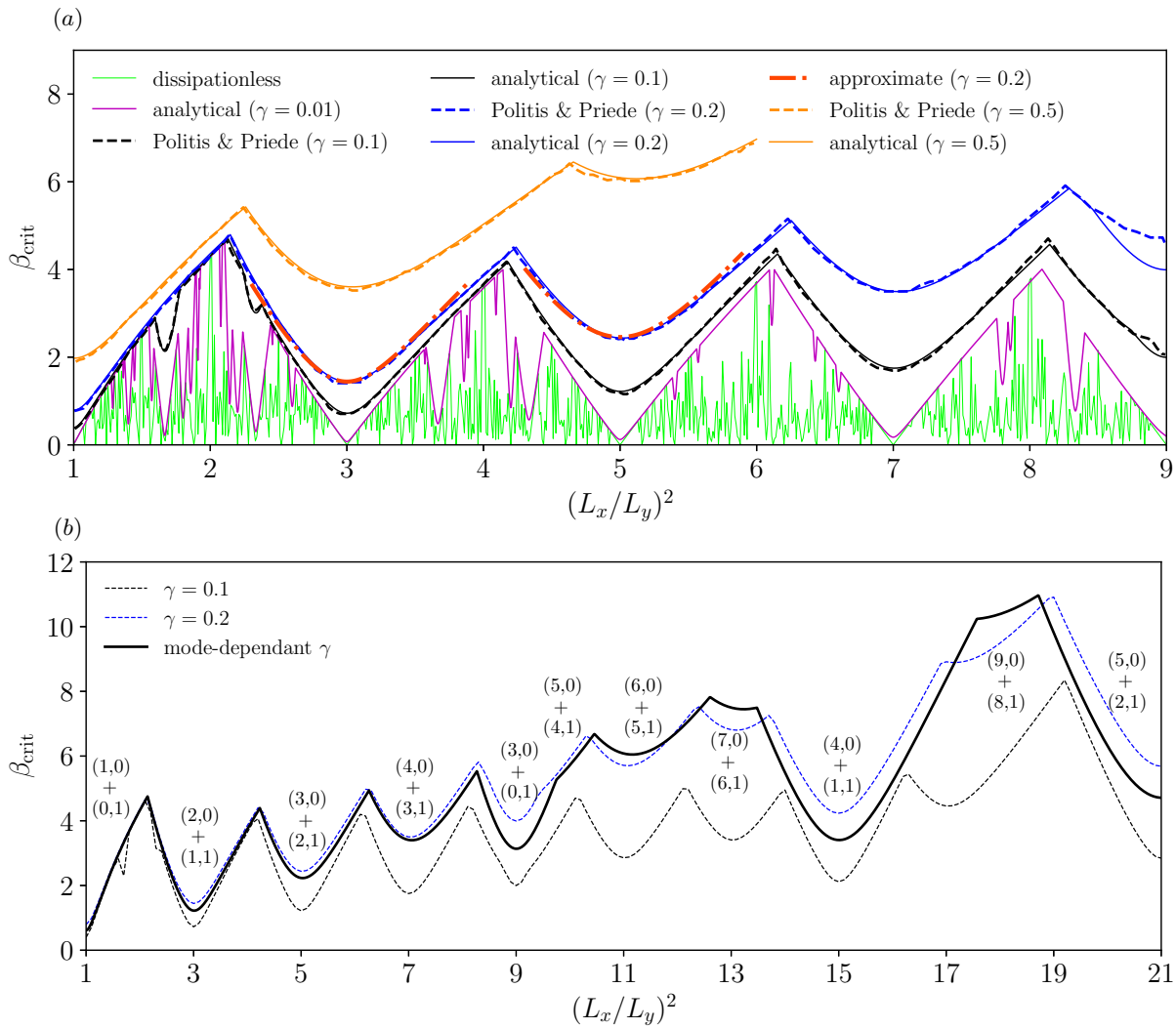

FIG. 6. Critical Sele parameters $\beta_{\rm crit}$ from Politis and Priede [28] and our analytical prediction [Eq. (24)] as a function of the aspect ratio squared $(L_x/L_y)^2$, with and without considering the effects of damping here modeled by (a) arbitrary constant damping coefficients γ. The non-dimensional coefficients γ were re-dimensionalized as $\lambda = 0.5\gamma\, c/L_x$ and $\lambda' = 0.5\gamma\, c/L_y$. The red curves show the approximate solution [Eq. (48)] for $\gamma = 0.2$ around $L_x/L_y = \sqrt{3}$ and $L_x/L_y = \sqrt{5}$. In (b) the instability onsets are plotted for mode-dependent friction coefficients calculated from damping rates in Eq. (39) for aspect ratios squared extending till 21. The solutions obtained from fixed coefficients are plotted as reference.

by using the mode-dependent viscous damping formulas, the change in the critical beta between different wave mode pairs can be more accurately described.

For these cases, a simple approximate solution for $\beta_{\rm crit}$ can be derived from our theory that is valid in the vicinity of the most unstable aspect ratios ($L_x^2/L_y^2 = 1, 3, 5, \dots$). Given these aspect ratios, we can set $N = 1$, because $L_x^2/L_y^2 = M/1$ with $M = 1, 3, 5, \dots$, and, by introducing Eq. (46) into (27), ignoring the term $i\delta\omega\delta\lambda$ and re-dimensionalising $\lambda = 0.5\gamma\, c/L_x$ and $\lambda' = 0.5\gamma\, c/L_y$, $\beta_{\rm crit}$ can be deduced as a function of only L_x/L_y and M:

$$\beta_{\rm crit} = \frac{\pi^4 \sqrt[4]{M}(M+1)}{16\sqrt{\delta_{M-1}}} \sqrt{\frac{\gamma^2}{\pi^2} + \frac{L_y}{2L_x}(M^2+1) + \frac{L_x}{L_y} - \sqrt{\frac{(M^2-1)^2}{4}\frac{L_y^2}{L_x^2} + (M+1)^2}}. \tag{48}$$

In this approximation, the numbers M must be taken as constants and the aspect ratio L_x/L_y as a variable. For $\gamma = 0.2$, we show Eq. (48) by the red curves for $M = 3$ and $M = 5$ in Fig. 6(a). The approximate solution is very accurate, but deviations become evident near the wave mode transition peaks so that the most stable aspect ratios cannot be determined analytically from Eq. (48). Onset values for the most unstable aspect ratios $L_x/L_y = 1, 3, 5, \dots$,

however, can readily be derived from Eq. (48) by setting $L_x/L_y = \sqrt{M}$ yielding the exact solution

$$\beta_{\text{crit}} = \frac{\pi^3 \sqrt[4]{M}(M+1)}{16\sqrt{\delta_{M-1}}}\gamma. \tag{49}$$

The physical relevance of the fractal distribution of critical aspect ratios, which is enveloped by the dissipative curves, has already been discussed in detail by Politis and Priede [28]. We make here a final note that the most unstable wave pairs (smallest β_{crit}) are not always the wave pairs with the highest growth rate. Therefore, other mode pairs than these predicted at $\beta_{\text{S}} = \beta_{\text{crit}}$ can develop at higher $\beta_{\text{S}} > \beta_{\text{crit}}$, which is why the exact pattern of MPR waves is difficult to predict even in the linear regime. At some certain point $\beta_{\text{S}} > \beta_{\text{crit}}$, it is always the primary rotating mode pair $(1,0)+(0,1)$ that yields the highest growth rate in all possible cell configurations, as already noted by Munger and Vincent [41]. But in these high β_{S} regimes, linear MPR models are often no longer applicable.

D. Comparison with measured stability onsets by Borisov *et al.* [21]

As Politis and Priede [28] have adopted constant damping parameters, it remains to compare our onset solution [Eq. (24)] in conjunction with wave mode-dependent damping rates (39), which constitute the major amelioration of our model. For this, the hitherto unrecognized model experiment by Borisov *et al.* [21] is the perfect (and only) option. The authors are the only ones to have ever succeeded in destabilizing MPR waves in two-layer experimental cells that can capture the complete MPR magnetohydrodynamics without any noteworthy trade-off. In their set-up, two transparent liquids were stratified (nitric acid ($HNO_{3(\text{aq})}$ at the bottom and $HNO_3 + C_5H_{12}O$ at the top), both of which had small but sufficiently contrasting electrical conductivities ($\sigma_1 = 0.67\,\text{S}\,\text{m}^{-1}$ and $\sigma_2 = 47.5\,\text{S}\,\text{m}^{-1}$). Due to the comparatively small difference in the density of the two liquids, small cell currents $I \sim 2-4\,\text{A}$, in combination with a strong external magnetic field $B_z = 0.25\,\text{T}$, were sufficient to attain large enough Sele parameters $\beta_{\text{S}} \sim 2$ to destabilize MPR waves. In order to change the electrical boundary conditions at the bottom electrode, the authors employed a removable insert made of porous material that was impregnated with an aqueous solution of nitric acid. This allowed both sets of idealized boundary conditions to be realized in the experiment, i.e. that the lower liquid be either a much better or much worse electrical conductor than the bottom electrode. Unfortunately, the authors have only published measurements without the insert, for which the ground electrode can be regarded as a perfect electrical conductor. This configuration is not accounted for by our model, but simply by changing the Neumann boundary condition [Eq. (5d)] to the Dirichlet boundary condition $\varphi_2 = 0|_{z=-h_2}$, we can easily adapt our solution. The stability onset has the same form as before:

$$\beta_{\text{crit}} = \frac{L_x^2 L_y^2}{h_1 h_2 \Theta}\sqrt{\frac{\bar{\lambda}^2 - \left[\mathrm{i}\delta\omega - \delta\lambda\right]^2}{(\bar{\omega}^2 - \delta\omega^2)\mathcal{T}_c\mathcal{T}_c'}}, \tag{50}$$

where,

$$\mathcal{T}_c = \frac{\Lambda}{k(k^2-k'^2)}\left[\frac{-k'}{\tanh[kh_1]} + \frac{k'\,\text{sech}[k'h_1]}{\sinh[kh_1]} + \frac{k}{\coth[k'h_1]} + \frac{-k'}{\tanh[kh_2]} + \frac{k'\,\text{sech}[k'h_2]}{\sinh[kh_2]} + \frac{k}{\coth[k'h_2]}\right],$$

$$\mathcal{T}_c' = \frac{\Lambda'}{k'(k'^2-k^2)}\left[\frac{-k}{\tanh[k'h_1]} + \frac{k\,\text{sech}[kh_1]}{\sinh[k'h_1]} + \frac{k'}{\coth[kh_1]} + \frac{-k}{\tanh[k'h_2]} + \frac{k\,\text{sech}[kh_2]}{\sinh[k'h_2]} + \frac{k'}{\coth[kh_2]}\right],$$

and the new conductivity jump parameter is

$$\Lambda = \frac{\sigma_1^{-1} - \sigma_2^{-1}}{\sigma_1^{-1}\tanh[kh_1] + \sigma_2^{-1}\tanh[kh_2]}.$$

Here, Borisov *et al.* [21] measured the critical onsets using a different dimensionless parameter $W = I_0 B_z/[(\rho_2 - \rho_1)gL_xL_y]$ that can easily be rewritten as β_{S} through $\beta_{\text{S}} = W\,L_xL_yh_1^{-1}h_2^{-1}$. In Fig. 7(a), the measured critical Sele parameters are shown together with the most unstable wave pairs $(m,n)+(m',n')$ in 9 different cells of different aspect ratios L_x/L_y in Fig. 7(a). For a better understanding, Fig. 7(b) further shows our continuous solution [Eq. (50)] compared with the stability prediction from a linear non-shallow MPR theory developed by Borisov *et al.* [42], who used the Galerkin method to formulate a Cauchy problem that was solved numerically. Fig. 7(c) supplements Fig. 7(b)

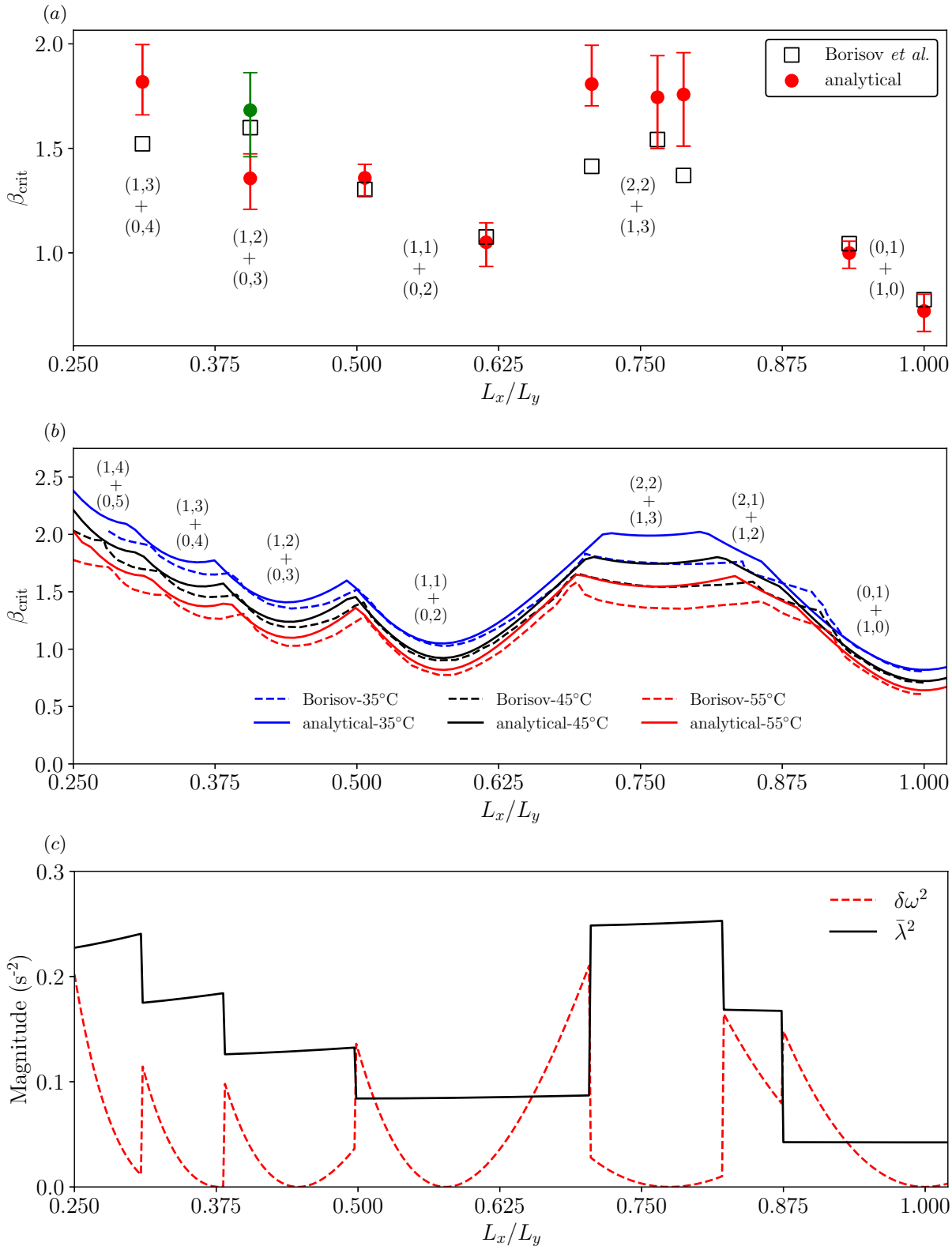

FIG. 7. (a) Comparison with the experimentally observed stability onsets at nine different cell geometries. The error bars cover a temperature range of 35°C to 55°C. (b) shows the analytical stability onset comparisons along with the critical wave modes observed at different cell aspect ratios. (c) is the distribution of parameters $\delta\omega^2$ and $\bar{\lambda}^2$ which play a key role in determining the critical wave mode and therefore, the critical onset value.

and shows the proportion of the two key parameters $\bar{\lambda}^2$ and $\delta\omega^2$ in Eq. (24), allowing the reader to determine whether a given onset value β_{crit} is influenced more by total damping or frequency mismatching. During their experiments, Borisov *et al.* [21] had to contend with the difficulty that the cell was subject to strong temperature fluctuations due to the high Ohmic losses (Joule heating) occurring in both poorly conducting liquids. The authors measured temperatures between $35° - 55°$C, which we incorporated in Fig. 7(a) using error bars. At all examined aspect ratios the reported wave pairs coincide with the most unstable pairs predicted by our model. Quantitatively, we also find fair agreement, but there are some considerable deviations, especially for the mode pair $(2,2)+(1,3)$. The measurement point that corresponds to $(1,2)+(0,3)$ lies very close to the jump point between the most unstable pairs $(1,2)+(0,3)$ and $(1,3)+(0,4)$, see Fig. 7(c). If we pick the neighbouring pair $(1,3)+(0,4)$ (green measuring point) here, although our model still predicts $(1,2)+(0,3)$ to be most unstable, we also find good agreement within the scope of measurement accuracies. In general, it can be observed that those measurements with the highest deviations also differ the most from Borisov's theory. The direct comparison of both theories reveals that our model always yields larger β_{crit}. Due to difficulties arising from translation, the reason for this discrepancy is quite difficult to identify, but we noticed that Borisov *et al.* [21] have entirely neglected sidewall damping. This is a misconception for non-shallow cells, we have estimated from our damping rates in Eqs. (40), (41) and (42) that sidewall damping accounts for more than a third of the total damping. Despite this stark difference, both theories concur remarkably well.

IV. CONCLUDING REMARKS

We present a perturbation formulation of the linear MPR instability, which originates from the superposition of two transverse standing wave modes, in the context of lab-scale reduction cell and LMB experiments. This is done by overcoming the shallow-water approximation and taking capillary wave regimes into account. The most important refinement, however, is that we consider dissipation not as before in terms of purely phenomenological damping parameters, but explicitly calculate the wave mode-dependent viscous damping rates. Because the resultant dissipation can differ considerably from wave mode to wave mode, see Fig. 7(c), this makes a sizable difference to the global stability.

All the analytical solutions described in this work are explicit so that the solutions can be implemented with minimal effort and without a wealth of prior knowledge. The main result of our study is provided by Eq. (22), which predicts MPR growth rates α_{MPR} for any given pair of standing wave modes. In a simple way, the global stability behavior is contained in this single equation: whenever the square root in Eq. (22) yields an imaginary number, the cell remains stable and only the natural frequencies of the waves are shifted by Lorentz forces. If the real part of the square root turns nonzero but the real part of α_{MPR} is negative, excited waves still attenuate, but more slowly than they would without MHD coupling, and both waves oscillate at the same frequency. Once the real part of α_{MPR} finally becomes positive, the cell becomes MPR unstable and an exponentially growing rotating wave motion builds up at the interface. The growth rate α_{MPR} incorporates the individual damping rates λ and λ' of the two coupled standing waves, which we have calculated separately.

To accurately determine the growth rate, it is imperative to quantify the viscous effects that play a large role in small-scale MPR systems. We have calculated the viscous damping by solving the leading-order Stokes boundary layers forming at all container walls as well as above and below the interface, and evaluated the Lamb's dissipation integral. The total damping rate λ [Eq. (39)] is composed of a wall damping term [Eq. (40)], an interfacial damping term [Eq. (41)] and a term resulting from the irrotational flow [Eq. (42)]. Independent of the MPR instability, these damping rates can be used for other problems, such as two-liquid sloshing experiments, and can further serve as a quick benchmark for direct numerical solvers. As demonstrated in Sec. III A, waves of different modes were simulated to determine the rate of amplitude decay in cell geometries of length 4,6,12 and 18 cm, akin to small-scale models. Performing the simulations for different aspect ratios and layer heights proved to be beneficial in two ways: firstly, it helped in verifying if the viscous boundary layers were accurately resolved, and secondly, in limit testing of the perturbative approach. The decay rates in cells of length $L_x = 4$ and 6 cm had notable differences from the theoretical values. This is mainly because the layer heights are comparable to the length of the viscous sub-layer (constituting more than 10% of the layer height), which violates our modeling to assume an irrotational bulk flow. Numerical damping in the larger cases with $L_x = 12$ and 18 cm, comparable to the geometry of most lab-scale experimental models, showed good agreement with theory. Due to the higher computational requirements associated with large grid sizes, numerical simulations were limited to cells with $L_x = 18$ cm.

The approach to calculate the mode-dependent viscous damping aided in determining precise instability onsets and critical wave modes as shown in Fig. 6(b). However, the complex inter-dependencies between the wave modes and the functions of the growth rate do not allow a straightforward approach to determine either the most volatile wave mode or the most unstable wave mode. This can be calculated only for inviscid cases under the shallow water approximation, see Sec. III C. Therefore, it is necessary to calculate the most unstable wave mode pair, which defines

TABLE II. Grid sizes adopted for different simulation cases in ARCHER. Cell sizes were kept uniform throughout the domain.

Aspect ratio	Length of cell L_x (cm)	Width of cell L_y (cm)	Layer height $h_1 = h_2$ (cm)	Grid size ($n_x \times n_y \times n_z$)
1	4,6,12 & 18	4,6,12 & 18	$L_x/8$	$64 \times 64 \times 16$ $128 \times 128 \times 32$ $256 \times 256 \times 64$
2		2,3,6 & 9	$L_x/8$	$64 \times 32 \times 16$ $128 \times 64 \times 32$
3		1.33,1,2 & 6	$L_x/8$	$192 \times 64 \times 48$

the critical stability parameter β_{crit} in Eq. (24), iteratively. To simplify the calculations for the reader, we have gathered all the essential equations with explanations for easy implementation, as well as a Python module, in the supplementary material.

One of the primary intentions of this stability theory is to calibrate our recent single-layer MPR experiment [24] in such a way that it can properly reflect the magnetohydrodynamics of (idealized) aluminum reduction cells. Doing so will allow us to measure and benchmark MPR growth rates of individual wave pairs. Moreover, our theory is well-positioned for augmenting the cell current (or external magnetic field) with alternating components, with a view to studying active MPR suppression techniques that have recently proved promising [43]. In Horstmann *et al.* [44] it was shown that the alternating current MPR suppression rests in a special form of parametric anti-resonance. However, the analysis relied on a highly idealized pendulum model with only two degrees of freedom. The ideal AC frequencies for a minimally invasive MPR suppression predicted in [44] still need to be confirmed by continuous MPR theories.

Appendix A: Additional details regarding the numerical analysis

Table II provides the mesh sizes adopted for all the cases simulated. The grid sizes were taken as a multiple of 2^n for grid independence tests. The mesh behaviour regarding damping rates and resonant frequency for different cell sizes of aspect ratio 1 is shown in Fig. 8(a) and 8(b). The values for all cases were found to converge at n_z=64, 32, and 48 for aspect ratios 1,2 and 3, respectively. Moreover, the same cases were repeated for different initial wave amplitudes as shown in Fig. 8(c), to verify that the amplitude was in no way influencing the resultant damping rates, similar to the theoretical values. The simulations were carried out with an initial amplitude of $L_z/500$. The damping rates remained unchanged when the cases were simulated with and without the advection term $\boldsymbol{\nabla} \cdot \rho \boldsymbol{u} \otimes \boldsymbol{u}$, confirming that the wave motion was within the scope of Stokes's flow. The total time of the simulation was at least 20 times the theoretical period of the oscillations. The wave frequencies and decay rates were extracted as shown in Fig. 8(d): by tracking the elevation of the interface at a specific point $(-L_x/2, -L_y/2)$, giving the local amplitude profile. The profile obtained was then fitted to an exponential curve, with the slope being the viscous damping rate. It can also be obtained by taking the volume integral of the kinetic energy $\int_{V_i} 0.5\rho_i U^2$, which decays twice as fast as the amplitude. The complete list of the frequencies and damping rates obtained from the simulations, in comparison with their respective theoretical values are given in Table III. Overall, we found good agreement with the theoretical values, especially in larger cells ($L_x = 12, 18$ cm).

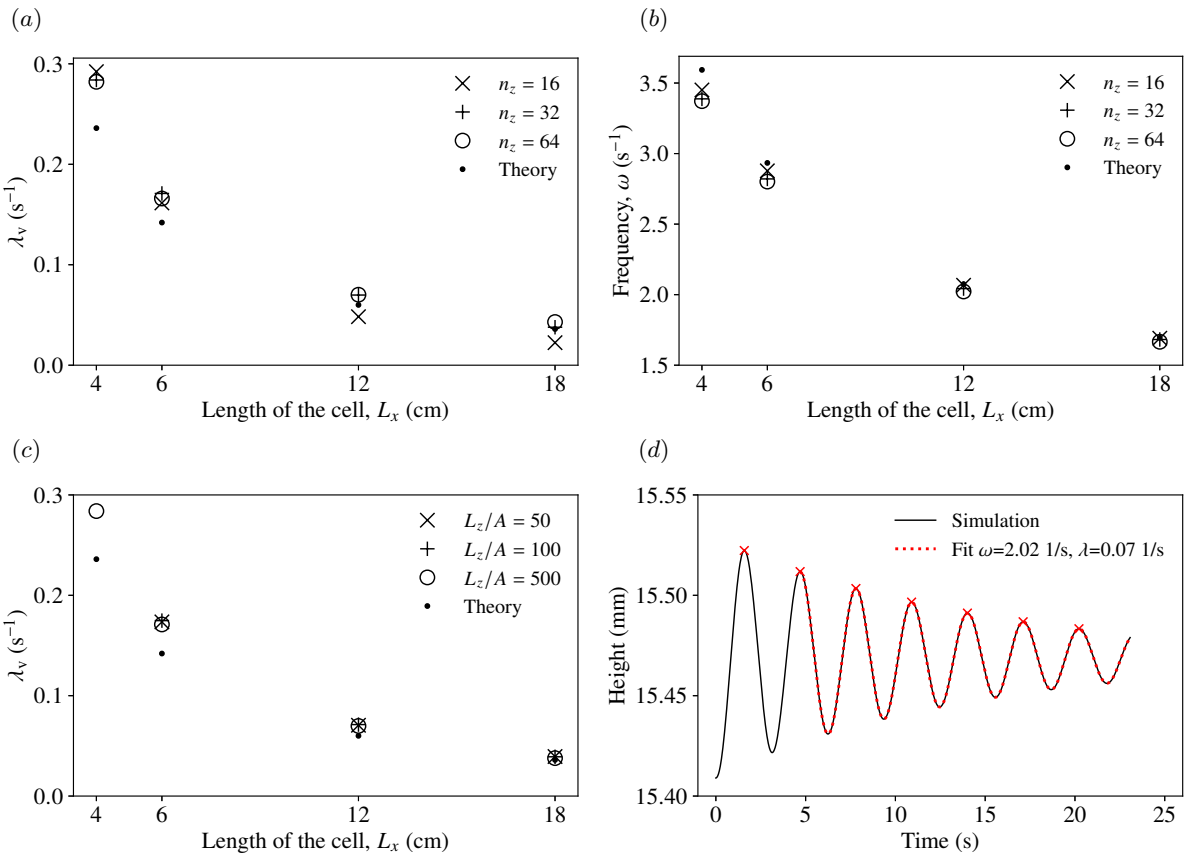

FIG. 8. The grid independence test observing the (a) viscous damping rate and (b) wave frequency for different cell sizes for aspect ratio 1 and wave number $(1, 0)$. (c) The same case was simulated with different initial wave amplitudes to certify the independence of initial wave amplitude on the damping rate. (d) the post-processing done on a cell size of $12 \times 12 \times 3$ cm, the resultant slope of the exponential curve fit of the amplitude signal at a local point $-L_x/2, -L_y/2$ to determine the wave frequency and the damping rate.

TABLE III. The detailed comparison of theoretical values of two-layer wave frequency and viscous damping with the numerical simulations of ARCHER code, for different cell dimensions and resultant wave modes. The two layers considered are cryolite and aluminum (see Table I). The heights of the respective layers are equal i.e. $h_1 = h_2$.

Aspect ratio	Cell dimensions, $L_x \times L_y \times [h_1 + h_2]$ (cm)	Wave mode (m,n)	Wave frequency ω (s^{-1})		Viscous damping rate λ (s^{-1})	
			Theory	ARCHER	Theory	ARCHER
1.0	4×4×1	(1,0)	3.593	3.372	0.236	0.282
	6×6×1.5		2.934	2.801	0.142	0.166
	12×12×3		2.075	2.022	0.060	0.070
	18×18×4.5		1.694	1.666	0.036	0.043
2.0	4×2×1	(2,0)	6.732	6.453	0.359	0.437
		(1,1)	7.383	7.078	0.403	0.504
	6×3×1.5	(2,0)	5.496	5.357	0.216	0.260
		(1,1)	6.028	5.879	0.243	0.296
	12×6×3	(2,0)	3.887	3.858	0.090	0.095
		(1,1)	4.262	4.233	0.102	0.104
	18×9×4.5	(2,0)	3.173	3.164	0.054	0.047
		(1,1)	3.480	3.471	0.061	0.051
3.0	4×1.34×1	(3,0)	9.258	8.875	0.474	0.585
		(0,1)	9.258	8.929	0.435	0.527
		(5,0)	12.89	12.44	0.644	0.819
		(4,1)	12.89	12.33	0.755	0.901
	6×2×1.5	(3,0)	7.559	7.347	0.285	0.347
		(0,1)	7.559	7.379	0.261	0.315
		(5,0)	10.52	10.28	0.385	0.485
		(4,1)	10.52	10.23	0.452	0.548
	12×4×3	(3,0)	5.345	5.288	0.119	0.137
		(0,1)	5.345	5.296	0.109	0.126
		(5,0)	7.440	7.384	0.161	0.177
		(4,1)	7.440	7.373	0.189	0.207
	18×6×4.5	(3,0)	4.364	4.343	0.072	0.073
		(0,1)	4.364	4.346	0.066	0.067
		(5,0)	6.075	6.054	0.097	0.089
		(4,1)	6.075	6.051	0.114	0.104

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